

Efficient Monkey Pox Diagnosis via Attention-Based MobileNetV2 Architecture

A Project Report submitted in the partial fulfillment of the Requirements for the award of the degree

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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KOTAPPAKONDA ROAD, YALAMANDA VILLAGE, NARASARAOPET- 522601

2024-2025

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CERTIFICATE

This is to certify that the project that is entitled with the name **“Efficient Monkey pox Diagnosis via Attention-Based MobileNetV2 Architecture”** is a bonafide work done by Shaik.Khaja Muneer (22475A0507), S.Srikanth (22475A0502), S.Vamsi (21471A0509) in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in the Department of COMPUTER SCIENCE AND ENGINEERING during 2024-2025.

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ACKNOWLEDGEMENT

We wish to express my thanks to carious personalities who are responsible for the completion of the project. We are extremely thankful to our beloved chairman sri **M. V. Koteswara Rao**, B.Sc., who took keen interest in us in every effort throughout thiscourse. We owe out sincere gratitude to our beloved principal **Dr. S. Venkateswarlu**, Ph.D., for showing his kind attention and valuable guidance throughout the course.

We express our deep felt gratitude towards **Dr. S. N. Tirumala Rao**, M.Tech., Ph.D., HOD of CSE department and also to our guide **M. Suneetha**, M.Tech , of CSE department whose valuable guidance and unstinting encouragement enable us to accomplish our project successfully in time.

We extend our sincere thanks towards **Dodda Venkata Reddy**, M.Tech., (Ph.D)., Assistant professor & Project coordinator of the project for extending his encouragement. Their profound knowledge and willingness have been a constant source of inspiration for us throughout this project work.

We extend our sincere thanks to all other teaching and non-teaching staff to department for their cooperation and encouragement during our B.Tech degree.

We have no words to acknowledge the warm affection, constant inspiration and encouragement that we received from our parents.

We affectionately acknowledge the encouragement received from our friends and those who involved in giving valuable suggestions had clarifying out doubts which had really helped us in successfully completing our project.

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DECLARATION

We declare that this project work titled " EFFICIENT MONKEYPOX DIAGNOSIS VIA ATTENTION-BASED MOBILENETV2 ARCHITECTURE" is composed by ourselves that the work contain here is our own except where explicitly stated otherwise in the text and that this work has been not submitted for any other degree or professional qualification except as specified.

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11. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Project Course Outcomes (CO'S):

CO421.1: Analyze the System of Examinations and identify the problem.

CO421.2: Identify and classify the requirements. **CO421.3:** Review the Related Literature

CO421.4: Design and Modularize the project

CO421.5: Construct, Integrate, Test and Implement the Project.

CO421.6: Prepare the project Documentation and present the Report using appropriate method.

Course Outcomes – Program Outcomes mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C421.1		✓											✓		
C421.2	✓		✓		✓								✓		
C421.3				✓		✓	✓	✓					✓		
C421.4			✓			✓	✓	✓					✓	✓	
C421.5					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
C421.6									✓	✓	✓		✓	✓	

Course Outcomes – Program Outcome correlation

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C421.1	2	3											2		
C421.2			2		3								2		
C421.3				2		2	3	3					2		
C421.4			2			1	1	2					3	2	
C421.5					3	3	3	2	3	2	2	1	3	2	1
C421.6									3	2	1		2	3	

Note: The values in the above table represent the level of correlation between CO's and PO's:

1. Low level
2. Medium level
3. High level

Project mapping with various courses of Curriculum with Attained PO's:

Name of the course from which principles are applied in this project	Description of the device	Attained PO
C2204.2, C22L3.2	The Monkeypox Diagnosis System uses Attention-Based MobileNetV2 for fast, accurate detection. Designed for mobile and edge devices, it enhances transparency with Grad-CAM and LIME. Future improvements include dataset expansion and real-time monitoring, advancing AI-driven healthcare.	PO1, PO3
CC421.1, C2204.3, C22L3.2	Each and every requirement is critically analyzed, the process model is identified	PO2, PO3
CC421.2, C2204.2, C22L3.3	Logical design is done by using the unified modelling language which involves individual team work	PO3, PO5, PO9
CC421.3, C2204.3, C22L3.2	Each and every module is tested, integrated, and evaluated in our project	PO1, PO5
CC421.4, C2204.4, C22L3.2	Documentation is done by all our four members in the form of a group	PO10
CC421.5, C2204.2, C22L3.3	Each and every phase of the work in group is presented periodically	PO10, PO11
C2202.2, C2203.3, C1206.3, C3204.3, C4110.2	Implementation is done and the project will be handled by the Doctors and in future updates in our project can be done based on detection of Monkeypox.	PO4, PO7
C32SC4.3	The physical design includes webpage to check whether an image is real or fake	PO5, PO6

ABSTRACT

The study proposed a deep learning-based solution for the accurate diagnosis of monkey pox based on attention enhanced MobileNetV2. Its focus was to sidestep these overlaps of symptoms that resonate with those of diseases characterized by chickenpox and measles. The structure was lightweight and thus could be used directly on smartphones and tablets, hence suitable for low-resource settings. The attention mechanism integrates spatial and channel attention, focusing the model's attention to the most critical regions in images along with the important features leading to detection accuracy. Because of limited images of monkey pox in the Monkey Pox Skin Images Dataset (MSID), images for chickenpox and measles were added to increase richness in the Models ability to distinguish between the diseases. An attention-based MobileNetV2 took and processed higher accuracy than models ResNet, VGG, and AlexNet taking 98.78% on extended MSID and 92% on original MSID. For model interpretability, Grad CAM and LIME techniques were used to show which parts of the images affected the model's decision, helping to establish trust with healthcare practitioners. In brief, such attention-augmented MobileNetV2 offers an applicable and high-accuracy solution for early detection in resource limited environments with added interpretability on mobile devices.

Efficient Monkey Pox Diagnosis via Attention-Based MobileNetV2 Architecture

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1. INTRODUCTION

1.1 Introduction

It is a viral infection caused by the Monkey pox virus under the genus of Orthopox viruses; it is closely related to the smallpox. It was first identified in monkeys in 1958, but then its existence was confirmed in the human population in the Democratic Republic of Congo in 1970. Its original infection points were central and western Africa, but it is rapidly spreading to become an emerging global issue as it spreads at a high rate into non endemic regions. As its name suggests, monkey pox is an infection of viral origin that can be transferred directly from animals to humans. There is direct transposition from person to person and by contaminated objects or fomites, thus human-to-human transmission can be risky during outbreaks [1].

The symptoms of monkey pox do not have specific characteristics other diseases that include chickenpox and measles; thus, it is challenging to diagnose even before the typical signs that include fever, swollen lymph nodes, and body aches followed by a rash developing into pustules. It also bears similarity with other forms of skin diseases that sometimes cause confusion and even misdiagnosis. The disease is fatal, ranging from 1% to 10% death rate depending on the strain; however, upon early detection, it can be controlled and outbreaks managed [2].

Traditional diagnosis methods, such as PCR tests, are reliable but mainly time-consuming and unfeasible in low resource settings. Early detection is the key for the prevention and successful treatment of monkey pox. Researchers have been looking into alternative solutions to provide early detection of monkey pox by applying deep learning models that will give a faster and more accessible method for diagnosis. In this paper, a model that efficiently detects monkey pox by using a deep learning model in developing a model based on MobileNetV2 specifically designed for resource-constrained settings [3].

Recent advancements in deep learning and artificial intelligence (AI) have paved the way for novel diagnostic approaches. MobileNetV2, a lightweight deep learning model, has emerged as a promising solution for disease classification due to its efficiency and capability to operate on mobile devices with minimal computational power. This model, when integrated with spatial and channel attention mechanisms, enhances its ability to focus on critical features in medical images, thereby improving classification accuracy [4].

The use of deep learning models in healthcare, particularly for diagnosing skin diseases, has gained traction. Traditional models such as VGG-16, ResNet-50, and InceptionV3 have been applied in classifying monkeypox from skin lesion images, achieving varied accuracy levels. However, many of these models suffer from poor generalizability due to limited datasets and the similarity of symptoms across different diseases. Moreover, explainability and interpretability of AI-driven models remain critical challenges in gaining the trust of healthcare professionals [5].

This study utilizes an extended version of the Monkeypox Skin Images Dataset (MSID), incorporating images of other skin conditions such as chickenpox, measles, eczema, lupus, molluscum, and scabies. By expanding the dataset, the model is trained to differentiate between diseases with similar visual characteristics, improving its diagnostic accuracy. The dataset undergoes preprocessing techniques such as resizing, normalization, and class balancing, ensuring optimal model performance [6]. This research aims to develop a robust, interpretable, and accessible deep learning model for monkeypox diagnosis. The proposed MobileNetV2-based approach is designed to operate in resource-constrained environments, making it a viable tool for early disease detection. By integrating AI-driven solutions with medical diagnostics, this study contributes to improving public health outcomes by enabling rapid and accurate disease identification, particularly in areas with limited healthcare infrastructure [7].

It highlights that Monkeypox is similar to smallpox, causing severe symptoms such as fever and pox-like lesions, which can lead to serious complications, morbidity, and economic burdens on healthcare systems. The introduction references the history of Monkeypox, noting its first reported case in 1958 and providing context with statistics on recorded global infections, particularly the cases in the United States [8].

the study "Monkeypox Diagnosis With Interpretable Deep Learning" discusses the rising prevalence of Monkeypox and the anticipated spread of a new strain in the coming years, as predicted by the Centres for Disease Control and Prevention (CDC) . The authors highlight the need for quick and automated detection methods for Monkeypox skin lesions, especially in regions with limited access to healthcare. They note that while deep learning (DL) techniques have been employed to detect various skin diseases, existing studies lack sufficient reasoning for their high prediction rates and do not incorporate generalization or regularization approaches to combat overfitting in DL models . This context sets the stage for the research, which aims to

address these gaps through the proposed method. the authors emphasize that the rapid spread of Monkeypox necessitates enhanced diagnostic tools that can effectively identify the disease through image analysis, particularly in underserved areas . They argue that while traditional diagnostic methods can be time-consuming and require significant expertise, the integration of deep learning technology could potentially streamline the diagnosis process, thus improving patient outcomes and management of the disease [9].

The text further critiques existing approaches in the field, observing that many deep learning models have achieved high accuracy rates but often lack transparency regarding the rationale behind their predictions. This opacity makes it challenging for healthcare professionals to trust and apply these systems in clinical settings . The study proposes the use of interpretable deep learning methods, specifically Local Interpretable Model-Agnostic Explanations (LIME), to provide insights into the model's decision-making process, thereby enhancing trust and usability in real-world applications .The introduction establishes the study's motivation and the critical need for innovative solutions, thereby laying a solid foundation for its objectives and methodology, which seeks to advance the capability of machine learning models in diagnosing Monkeypox lesions effectively [10].

Monkeypox is an infectious disease caused by the Monkeypox virus, which belongs to the Orthopoxvirus family. Its symptoms closely resemble those of smallpox, making early diagnosis crucial to controlling outbreaks. Since the eradication of smallpox in 1980 and the discontinuation of smallpox vaccinations, monkeypox has become a significant global health concern. Though originally endemic to Central and West Africa, recent outbreaks in non-endemic regions have heightened the need for rapid and accurate detection methods [11].

Traditional diagnosis methods, such as polymerase chain reaction (PCR), are effective but require extensive laboratory resources, making them inaccessible in low-resource settings. To address these challenges, researchers have turned to artificial intelligence (AI) and deep learning models for automated disease detection. However, training deep learning models requires a substantial dataset, which is limited in the case of monkeypox To overcome this limitation, this study proposes a **federated deep learning framework** that leverages **CycleGAN-based data augmentation** and secure **federated learning (FL)** techniques. The framework integrates deep learning models such as **MobileNetV2**, **Vision Transformer (ViT-B32)**, and **ResNet50** to classify monkeypox skin lesions accurately while ensuring data privacy.

The proposed system provides a privacy-preserving, efficient, and accurate monkeypox detection method that can be deployed in real-world healthcare environments. By combining AI-driven diagnosis with secure data-sharing mechanisms, this research contributes to improving public health by enabling early disease identification and preventing the spread of monkeypox.[12].

Monkeypox is a viral disease caused by the Monkeypox virus (MPXV), which belongs to the Orthopoxvirus family, the same family of viruses that causes smallpox. Due to its sporadic outbreaks, the World Health Organization (WHO) has classified monkeypox as a global health emergency. The disease, first discovered in 1958, has rapidly spread across 73 countries, with over 65,000 reported cases as of September 2022. The most common method for diagnosing monkeypox is the polymerase chain reaction (PCR) test, but early detection remains a challenge[13].

In recent years, social media platforms such as Twitter have played a crucial role in public discourse, influencing opinions and shaping perceptions of global health crises. Social media has become an essential tool for monitoring public sentiment, misinformation, and panic associated with outbreaks. Analyzing sentiments expressed on Twitter provides valuable insights into public reactions and helps policymakers craft better health communication strategies. This study aims to conduct a machine learning-based sentiment analysis on public opinions regarding the monkeypox outbreak using Twitter data. The research involves collecting over 500,000 multilingual tweets, processing them using natural language processing (NLP) techniques, and applying machine learning models to classify sentiments as positive, negative, or neutral. The study explores various text preprocessing methods, vectorization techniques like CountVectorizer and TF-IDF, and multiple classification algorithms such as Support Vector Machines (SVM), Random Forest, Logistic Regression, and Neural Networks. The best-performing model is identified based on accuracy, precision, recall, and F1-score. By understanding public sentiment, this research contributes to effective health policy decision-making, improved public awareness, and proactive mitigation strategies to control the spread of misinformation and stigmatization. The findings will help health authorities and governments improve their communication strategies to manage public perception and response to future outbreaks [14][15].

2.LITERATURE SURVEY

The paper discusses several deep learning techniques used in the classification of Monkey pox based on skin Lesion images [9] [10]. In this regard, several models were applied for classifying skin conditions, including monkey pox [4]. For instance, research applied models such as VGG- 19, ResNet-152, InceptionV3, and even MobileNetV2 to the classification process of the skin conditions. These models, however suffered the disadvantage of limited dataset and number of classes, thus limiting generalizability. [2][7]

Most deep learning-based studies on skin lesion detection relied heavily on knowledge transfer to learn in regions where PCR testing was limited [5][6]. A good number of the models have shown some good accuracy in the automatic detection of skin lesions but still remain less robust in real-world applications since they treated fewer categories of skin diseases. One of them, PoxNet22, attained 100% accuracy, precision, and recall in classifying monkey pox but failed to have interpretability as the requirement for clinical diagnostics. PoxNet22's further binary classification also restricts it to analyze between mixed skin conditions. [1]

Another approach adopted here is to deal with the limitation of training data, that is, data augmentation. Although created images by data augmentation cannot express real medical conditions, it may impact the practical application model in an actual clinical environment. Models VGG-19, AlexNet, and ResNet-152 were doing very good jobs but lacked an explainability aspect, which is of utmost importance in giving transparency about the doctors' diagnostics [3][8]. In recent work, attention has been devoted to improving explanations of AI models with techniques like LIME and Grad-CAM, which can provide deeper insights that how the model actually makes a decision. This is helpful for building trust and transparency; in the proposed attention-based MobileNetV2 architecture, efficient, explainable monkey pox diagnosis will be achieved.

Several studies have demonstrated the effectiveness of sentiment analysis in tracking public perceptions during disease outbreaks. For example, during the COVID-19 pandemic, researchers leveraged Twitter data to analyze public sentiment using deep learning models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). Chakraborty et al. (2020) applied deep learning classifiers combined with Gaussian functions to classify sentiments from COVID-19-related tweets, emphasizing the importance of monitoring

misinformation to prevent public panic. Shahi et al. (2022) focused on sentiment analysis in the Nepali language, using TF-IDF and FastText for feature extraction and classification [8]. Their results highlighted the effectiveness of hybrid machine learning approaches in capturing public sentiment. Similarly, Chinnasamy et al. (2022) analyzed vaccine hesitancy using Twitter data, demonstrating that public attitudes toward vaccination evolved over time[8][5].

the study "Monkeypox Diagnosis With Interpretable Deep Learning" presents an in-depth exploration of the applications of artificial intelligence (AI) and deep learning (DL) techniques in the diagnosis of skin-related diseases. It begins by highlighting the growing adoption of deep learning in the medical field, particularly its effectiveness in detecting various skin diseases[1]. Numerous studies indicate that DL models are capable of achieving remarkable accuracy in image classification tasks, thereby demonstrating their potential in accurately identifying skin lesions associated with infections. However, the review also addresses significant challenges that arise from the implementation of these technologies. Despite many existing models achieving high accuracy rates, they often suffer from a lack of interpretability, which can inhibit trust and adoption among healthcare practitioners who require a clear understanding of how predictions are made in order to feel confident in relying on AI-driven solutions[10].

To address this issue, the literature emphasizes the urgent need for models that are not only precise but also interpretable. Techniques such as Local Interpretable Model-Agnostic Explanations (LIME) and SHapley Additive exPlanations (SHAP) are referenced as means to provide insights into the decision-making processes of complex models, thus tackling the black-box nature that commonly characterizes deep learning algorithms. Additionally, the review discusses the advantages of transfer learning — a technique that allows the use of pre-trained models to improve the efficiency of training on smaller, medical imaging datasets. By leveraging transfer learning, researchers can enhance model performance while mitigating the computational demands typically associated with training deep learning models from scratch [10][16].

Moreover, the authors underscore that despite numerous studies reporting high levels of predictive accuracy, there remains a significant scarcity of research confirming the generalizability and robustness of these models across diverse datasets and clinical scenarios. This lack of external validation raises concerns regarding the models' tendency to overfit to

training data, potentially leading to poor performance in real-world applications.[13][14] In summary, the literature review effectively documents the advancements in deep learning for medical diagnostics while also highlighting key areas necessitating improvement, particularly in the context of Monkeypox diagnosis. It successfully establishes the rationale for the authors' research, identifying the gaps in existing studies that their work aims to address, especially regarding the interpretability and reliability of deep learning models in healthcare settings.

Monkeypox transmission dynamics analyzes various existing models and research findings related to infectious diseases, particularly those affecting public health. It begins by underscoring the growing interest in mathematical modeling as a critical tool for understanding epidemic behaviors and informing effective health interventions. Several studies have previously examined the spread of Monkeypox, elucidating its epidemiological characteristics and transmission routes. Research highlighted by authors such as Fine et al. demonstrates the effectiveness of smallpox vaccinations against Monkeypox, showing an 85% efficacy [6]. Other significant works explore epidemic control strategies, focusing on treatment modalities and vaccination campaigns. In addition to Monkeypox, comparable models have been developed for various viral infections, such as the chikungunya virus, which incorporate vaccination and treatment strategies to mitigate spread. The review emphasizes that a robust framework combining awareness, quarantine, and vaccination is essential for developing targeted interventions, especially in urban settings prone to rapid disease transmission. Ultimately, this synthesis of prior research establishes a foundational understanding of how mathematical modeling can aid in confronting emerging zoonotic diseases, aligning with public health objectives to safeguard communities from outbreaks [16][17].

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

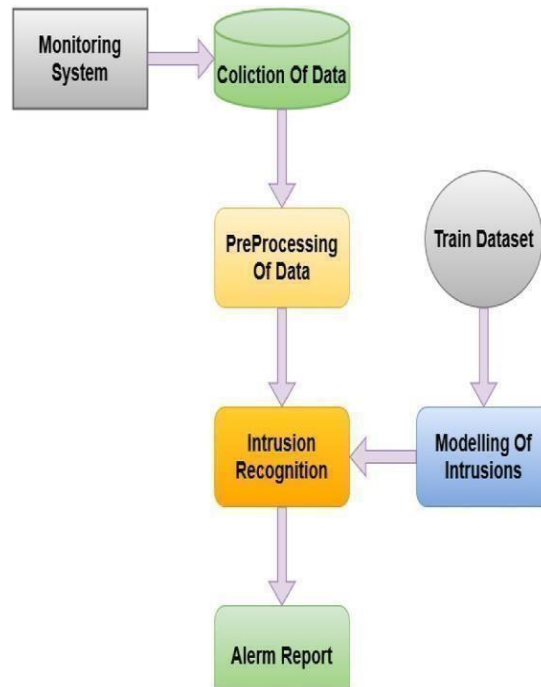


Fig 3.1 Flow chart of existing system

Intrusion Detection Systems (IDS) employ various techniques to detect unauthorized access or suspicious activity. Signature-based detection compares network activity against known attack patterns, effectively identifying familiar threats but struggling with new ones. In contrast, anomaly-based detection establishes a baseline of normal behavior, flagging deviations as potential threats, which allows it to detect unknown attacks but can lead to more false positives. Heuristic and behavioral-based methods analyze user and network actions to identify unusual behavior patterns. Increasingly, machine learning (ML) techniques—including supervised and unsupervised models—are leveraged to dynamically classify and adapt to new attack patterns. Advanced IDS often use hybrid detection, combining multiple methods, such as ML with signature-based approaches, enhancing accuracy while reducing false positives. Additionally, reputation-based detection assesses traffic based on the history of IP sources, adding a layer of quick threat identification. Together, these techniques provide a robust, adaptive defense against evolving cyber threats.

3.2 DISADVANTAGES OF THE EXISTING SYSTEM

Despite the advancements in deep learning models for monkeypox diagnosis, the existing systems face several limitations:

1. High Computational Requirements

- Traditional deep learning models such as ResNet-152, VGG-19, and AlexNet require significant computational resources, making them unsuitable for real-time applications on mobile or edge devices.
- These models struggle to operate efficiently in resource-constrained environments, such as rural healthcare centers.

2. Limited Interpretability

- Many existing models lack explainability, making it difficult for healthcare professionals to trust and validate AI-generated diagnoses.
- Methods like PoxNet22 achieved high classification accuracy but lacked interpretability, reducing their clinical applicability.

3. Data Limitations and Poor Generalization

- Most deep learning models rely on limited datasets that do not adequately represent diverse skin conditions.
- The models often fail to generalize well when presented with new or slightly varied monkeypox cases.

4. Challenges in Differentiating Similar Skin Diseases

- Monkeypox shares visual similarities with diseases like chickenpox, measles, and other dermatological conditions, leading to high misclassification rates in certain models.
- Without specialized attention mechanisms, existing models struggle to focus on subtle differentiating features.

5. Dependence on Pre-Trained Models

- Many deep learning approaches rely on transfer learning from models pre-trained on general image datasets, which may not be optimized for medical images.
- Fine-tuning these models for monkeypox diagnosis requires large datasets, which are often unavailable.

6. Lack of Real-World Deployment

- Most existing research focuses on model accuracy in controlled settings but lacks real-world validation and clinical deployment.
- The absence of mobile-friendly and edge-compatible solutions limits practical applications in remote areas.

7. Absence of Attention Mechanisms

- Standard CNN models fail to focus on the most relevant regions of an image, reducing diagnostic accuracy.
- Without spatial and channel attention, feature extraction remains suboptimal, leading to misinterpretations.

To address these drawbacks, the proposed system leverages attention-based MobileNetV2, which enhances feature extraction, improves interpretability with Grad-CAM and LIME, and ensures efficient operation on mobile and edge devices.

3.3 PROPOSED SYSTEM

To overcome the limitations of existing monkeypox diagnosis models, this study proposes an Attention-Based MobileNetV2 Architecture that enhances efficiency, interpretability, and accuracy while ensuring compatibility with mobile and edge devices. The proposed system integrates deep learning, attention mechanisms, and explainability techniques to provide a reliable, real-time, and accessible solution for monkeypox detection.

The proposed system adapts a multi-level ensemble learning approach combined with the transfer learning approach to promote the accuracy and efficiency for breast cancer diagnosis using an ultrasound imaging approach. Here, MLPs and SVMs are used as base classifiers which capture the different patterns in the datasets, and their strengths ensure robust classification. These outputs are aggregated through an ensemble stacking method that makes intelligent predictions to enhance performance. Further, the transfer learning enhanced the system to use a pre-trained model such as VGG19 for feature extraction to effectively capture high-level, domain-specific features with a smaller dataset. This integration of ensemble learning and transfer learning reduces false negatives while increasing sensitivity and specificity in differentiating between benign and malignant cases. By balancing the high diagnostic accuracy with the computational efficiency, the proposed system provides a scalable and adaptive solution for the detection of breast cancer by addressing the complexity and variability of medical imaging data.

Advantages:

1. Improved Detection Accuracy
2. Enhanced Adaptability to New Threats
3. Reduced False Positives

4. Efficient Resource Utilization
5. Real-Time Detection Capabilities
6. Scalability for Large-Scale Networks
7. Low Computational Overhead
8. Effective Knowledge Transfer Through Distillation

Flowchart of Proposed Monkeypox Diagnosis System

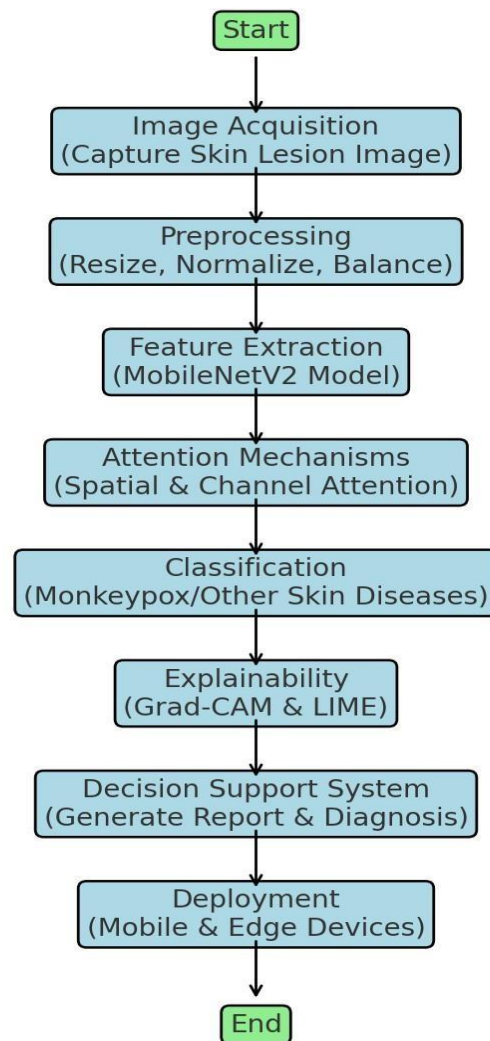


Fig 3.3 Flow chart of proposed system

The flowchart outlines the proposed system for machine learning, starting with Data Collection, where raw data is gathered. Next, Data Processing involves cleaning and organizing the data for model training. In Initial Model Training, a preliminary model is developed using the processed data. The system then undergoes Two-Level Ensemble Learning, where multiple models are combined to improve performance.

The proposed **Attention-Based MobileNetV2 Architecture** for monkeypox diagnosis follows a structured workflow to ensure efficient, accurate, and interpretable disease detection. The process begins with **image acquisition**, where a user, either a patient or a healthcare professional, captures a high-resolution image of the affected skin area using a mobile device or diagnostic tool. To enhance the model's performance, the image undergoes **preprocessing**, which includes resizing to 224×224 pixels, normalization to scale pixel values, and class balancing to ensure fair training across different skin disease categories.

Once preprocessed, the image is passed through **MobileNetV2 for feature extraction**, where key patterns such as lesion shape, texture, and color are identified. The system further improves classification accuracy by incorporating **attention mechanisms**, specifically spatial attention, which focuses on the most relevant lesion areas, and channel attention, which highlights important texture and color variations. After extracting features, the model classifies the input image into one of several categories, including **Monkeypox, Chickenpox, Measles, and other skin diseases such as Eczema, Lupus, Molluscum, and Scabies**. To ensure transparency and interpretability, the system integrates **explainability techniques** such as **Grad-CAM and LIME**, which provide visual explanations and simplified interpretations of model predictions. This step is crucial in building trust among healthcare professionals and allowing them to validate the AI's decisions. Following classification and explainability, the system generates a **detailed diagnostic report**, which includes the predicted disease category, confidence score, and recommendations for further medical consultation if necessary.

For accessibility and real-time application, the model is optimized for **deployment on mobile and edge devices**, allowing self-diagnosis through mobile apps and real-time analysis in hospitals and remote healthcare centers. Additionally, the system can be integrated into **telemedicine platforms**, providing remote diagnosis capabilities. The process concludes once the user receives the final diagnosis and recommendations. This structured approach ensures **fast and accurate disease detection**, supporting early intervention, reducing misclassification, and making AI-driven healthcare more accessible in low-resource settings.

3.4 FEASIBILITY STUDY

The feasibility study evaluates the practicality and viability of implementing the Attention- Based MobileNetV2 Architecture for Monkeypox Diagnosis. It assesses the system's effectiveness from technical, economic, and operational perspectives, ensuring that the proposed solution is realistic and beneficial for real-world deployment.

The study of the possibility takes into account the following four main considerations:

1. Economic Feasibility
2. Technical Feasibility
3. Operational Feasibility
4. Social and Ethical Feasibility

Economic Feasibility

From a financial perspective, the system is cost-effective as it reduces the reliance on expensive laboratory-based diagnostic methods such as PCR testing. Since MobileNetV2 is lightweight and optimized, it does not require expensive hardware, making it a budget-friendly solution for healthcare institutions and governments. Additionally, the use of pre-trained deep learning models and transfer learning reduces the cost and time needed for training from scratch. The system's ability to function on existing smartphones and edge devices eliminates the need for additional infrastructure, further enhancing economic feasibility.

Technical Feasibility

The proposed system is technically feasible as it leverages **MobileNetV2**, a lightweight deep learning model that is efficient and optimized for mobile and edge computing. The use of **attention mechanisms (spatial and channel attention)** enhances feature extraction, improving classification accuracy while keeping computational costs low. Additionally, **explainability techniques (Grad-CAM & LIME)** ensure transparency in AI-driven decisions, making it more acceptable for healthcare professionals. The system requires moderate computing power, making it deployable on **mobile devices, hospitals, and telemedicine platforms**, ensuring accessibility even in resource-limited settings.

Operational Feasibility

The system is designed for ease of use and integration into existing healthcare workflows. Patients and healthcare professionals can easily capture skin lesion images using a smartphone or medical device. The automated classification and decision support system

ensures that healthcare practitioners receive quick and reliable results, enabling early disease detection. The user-friendly interface and real-time feedback make it accessible even for non-technical users. Moreover, the ability to integrate the model with telemedicine platforms and hospital systems improves its operational efficiency, allowing widespread adoption.

Social and Ethical Feasibility

The system promotes equitable healthcare access, particularly in remote and underdeveloped areas where PCR testing is not readily available. By using AI-driven diagnosis, it helps reduce the burden on healthcare facilities and ensures faster detection of monkeypox outbreaks. Ethical considerations, such as data privacy and security, are addressed through the use of explainability techniques, ensuring transparency and trust in AI-based decision-making.

Importance of Feasibility Study

A feasibility study is essential in determining the practicality and viability of a project before investing significant resources. It evaluates technical feasibility by ensuring that the required technology, tools, and infrastructure are available for successful implementation. The economic feasibility aspect helps assess cost-effectiveness, ensuring that the project remains financially sustainable while reducing unnecessary expenses. Operational feasibility ensures that the system can be easily integrated into existing workflows, improving efficiency and usability for stakeholders. Additionally, a feasibility study helps identify risks and uncertainties, allowing for proactive mitigation strategies to avoid technical failures or financial losses. It also aids in decision-making by providing a structured analysis of the project's potential success. Finally, in healthcare applications like AI-driven monkeypox diagnosis, social and ethical feasibility is crucial to ensure patient data privacy, transparency, and equitable healthcare access. By addressing these factors, a feasibility study plays a critical role in ensuring the successful development and deployment of any system.

4. SYSTEM REQUIREMENT

4.1 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specifications (SRS) define the functional and non-functional requirements of a system, ensuring that all stakeholders understand its objectives, features, and constraints. For the Attention-Based MobileNetV2 Monkeypox Diagnosis System, the SRS outlines essential components such as image acquisition, preprocessing, feature extraction, classification, and explainability using AI-driven techniques like Grad-CAM and LIME. The system must be optimized for mobile and edge deployment, ensuring accessibility in low-resource settings. Key non-functional requirements include high accuracy ($\geq 95\%$), fast response time (2-5 seconds), data privacy, and usability across different platforms. The software is developed using Python, PyTorch/TensorFlow, Flask/Django, and cloud technologies like Google Cloud or AWS for scalable performance. By defining clear technical, security, and performance requirements, the SRS ensures the system is efficient, reliable, and practical for real-world healthcare applications.

Scope:

The scope of the Monkeypox Diagnosis System extends beyond basic image classification, offering real-time, interpretable, and mobile-friendly disease detection. The system enables patients, healthcare professionals, and researchers to detect Monkeypox, Chickenpox, Measles, and other skin diseases through an AI-driven deep learning model with attention mechanisms. It supports deployment on mobile devices, web applications, and hospital systems, making it scalable for telemedicine and remote healthcare access. Additionally, the system enhances public health efforts by allowing early detection and outbreak monitoring, reducing the burden on healthcare facilities. The project is designed to be cost-effective, user-friendly, and privacy-compliant, ensuring global accessibility and impact in both urban and rural settings.

4.2 REQUIREMENT ANALYSIS

Requirement analysis is a crucial phase in the development of the Attention-Based MobileNetV2 Monkeypox Diagnosis System, ensuring that all functional and non-functional needs are identified and addressed. This process involves gathering and evaluating requirements from healthcare professionals, patients, and researchers to build a reliable, efficient, and user-friendly system. Functional requirements include image acquisition, preprocessing, deep

functional requirements focus on accuracy, performance, security, and scalability. The analysis also considers hardware and software constraints, ensuring compatibility with mobile devices, web applications, and cloud platforms. By conducting a thorough requirement analysis, the system is designed to provide high accuracy, real-time processing, and seamless integration into telemedicine and hospital systems, making AI-driven monkeypox diagnosis accessible and effective for early disease detection.

Data Collection:

In this research, our dataset was primarily sourced from two premier databases: the Monkeypox Skin Images Dataset (MSID) and DermNet. These datasets can be found at <https://www.kaggle.com/datasets/dipuiucse/monkeypoxskinimagedataset>, <https://www.kaggle.com/datasets/shubhamgoel27/dermnet> Data collection is a critical step in developing the Attention-Based MobileNetV2 Monkeypox Diagnosis System, as the accuracy and reliability of the model depend on a well-curated dataset. The primary source of data is the Monkeypox Skin Images Dataset (MSID), which includes images categorized into Monkeypox, Chickenpox, Measles, and Normal Skin Conditions. To enhance the model's generalizability, the dataset is extended with additional skin conditions such as Eczema, Lupus, Molluscum, and Scabies from public dermatology repositories like DermNet. All images are collected from verified medical sources, research databases, and public health organizations to ensure high-quality and medically relevant data. The dataset is then preprocessed through image resizing (224×224 pixels), normalization (scaling pixel values between 0-1), and class balancing to improve model performance. By incorporating a diverse and well-labeled dataset, the system ensures better accuracy, reduced misclassification, and enhanced real-world applicability, making AI-driven monkeypox diagnosis more effective and reliable.

Data Pre-processing:

1. Image Resize:

All the images have been re-sized into the fixed resolution of 224 pixels' x 224 pixels uniformed over the dataset. This is an important step because it allows for uniformity in size inside the deep learning model and lets it process the images into one uniform size, which maintains the efficiency behind convolutional operations and precision.

2. Normalization:

Normalization was used as the scaling factor for all pixel values of images. RGB values of all pixels lay between 0 and 255. The given value for all the pixels was divided by 255 to normalize the value between 0 and 1.

3. Class Imbalance Handling:

Since this dataset was imbalanced in terms of number of images per class, the dataset was taken care of with careful stratified splitting so that both the training and test sets should contain proportionate presentations from each class of skin diseases, thus giving a balanced distribution without favoring any single class.

4. No Data Augmentation:

Unlike most of the image-based deep learning research which artificially expands the datasets through flipping, rotation or cropping, no data augmentation was done in the present study. This is because the authenticity of medical images has to be ensured for proper diagnosis in real-life scenarios. Thus, this study has relied on the training of the model using real, unaltered medical images.

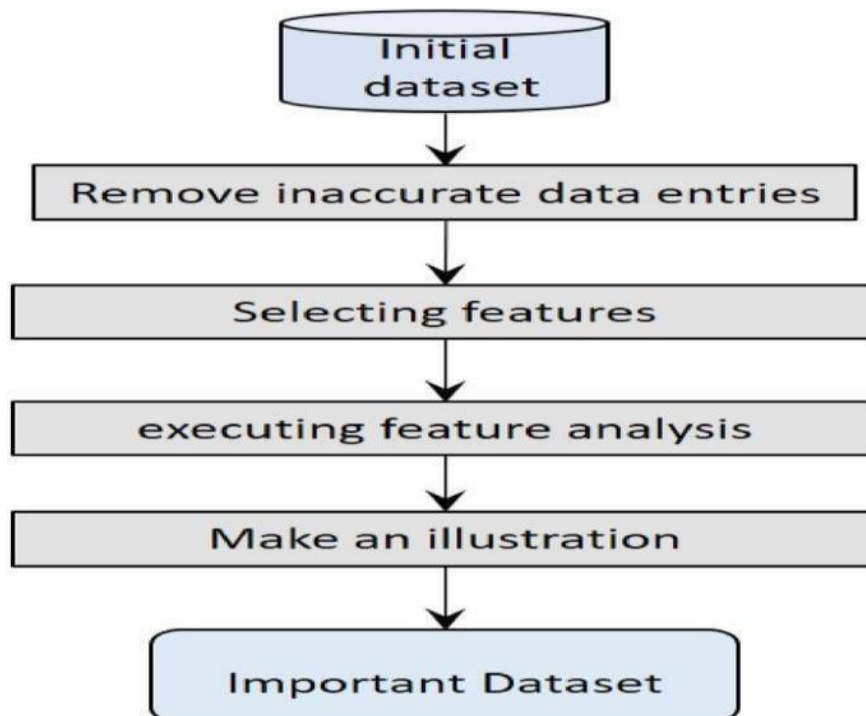


Fig 4.2 Data Pre-processing

Dataset Splitting:

Dataset splitting is a crucial step in training and evaluating the Attention-Based MobileNetV2 Monkeypox Diagnosis System, ensuring the model learns effectively while preventing overfitting. The dataset is typically divided into training, validation, and testing sets to optimize performance. In this system, 80% of the data is allocated for training, allowing the model to learn patterns and features, while 10% is used for validation, helping to fine-tune hyperparameters and avoid overfitting. The remaining 10% is reserved for testing, providing an unbiased evaluation of the model's real-world performance. This balanced approach ensures that the model generalizes well to unseen images, improving accuracy, robustness, and reliability when diagnosing monkeypox and similar skin conditions in diverse environments.

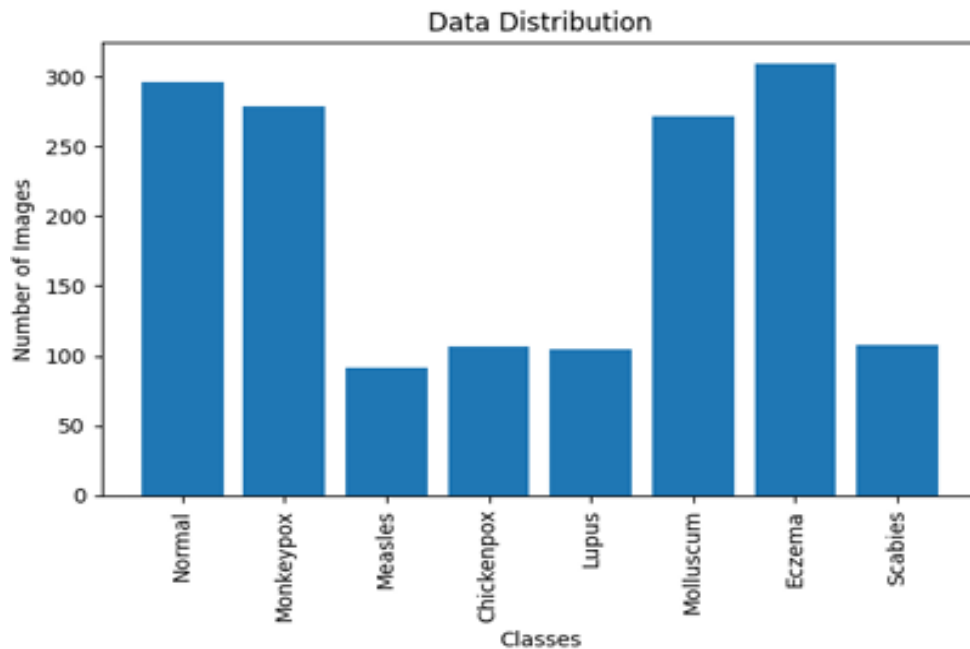


Fig 4.2 Data Distribution

Model Training:

The Attention-Based MobileNetV2 model is trained using a preprocessed dataset with transfer learning for faster convergence. The Adam optimizer, categorical cross-entropy loss, and early stopping are applied to improve accuracy and prevent overfitting. The model is trained in multiple epochs, refining its ability to classify Monkeypox, Chickenpox, Measles, and other skin conditions. Performance is evaluated using accuracy, precision, recall, and F1-score for real-world deployment.

Model Evaluation and Testing:

The trained Attention-Based MobileNetV2 model is evaluated using accuracy, precision, recall, F1-score, and specificity to ensure reliability. A testing dataset (10%) is used to assess real-world performance and generalization. Grad-CAM and LIME provide explainability, helping healthcare professionals interpret predictions. The model is validated for robustness and clinical applicability before deployment.

4.3 Hardware Requirements:

- System Type : intel@core™i3-7500UCPU@2.40gh
- Cache memory : 6MB(Megabyte)
- RAM : 8GB (gigabyte)
- Hard Disk : 500GB
- Graphic card : NVIDIA GPU (GTX 1050)

4.4 Software Requirements:

- Operating System : Windows 11, 64-bit Operating System
- Coding Language : Python
- Python distribution : Google Colab, Flask
- Browser : Any Latest Browser like Chrome

5. SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The Attention-Based MobileNetV2 Monkeypox Diagnosis System follows a multi-layered architecture for efficient disease detection. It consists of the following key components:

1. **Input Layer:** Users capture or upload skin lesion images through a mobile app or web interface.
2. **Preprocessing Layer:** Images undergo resizing (224×224 pixels), normalization, and class balancing to improve model accuracy.
3. **Feature Extraction Layer:** MobileNetV2 with spatial and channel attention mechanisms extracts key features for classification.
4. **Classification Layer:** The model categorizes images into Monkeypox, Chickenpox, Measles, or Other Skin Conditions using deep learning.
5. **Explainability Layer:** Grad-CAM and LIME provide visual explanations to enhance interpretability for healthcare professionals.
6. **Decision Support & Deployment:** The system generates diagnostic reports and is optimized for mobile, cloud, and edge device deployment.

This architecture ensures high accuracy, efficiency, and real-time disease detection, making it accessible in low-resource settings

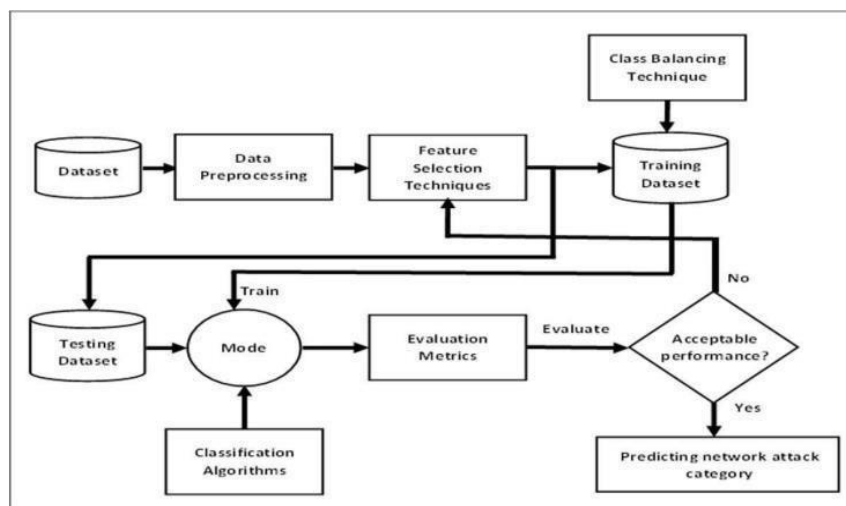


Fig 5.1 Flow chart of system Architecture

5.2 FEATURE EXTRACTION:

Feature extraction in the proposed system involves processes and mechanisms to derive meaningful patterns from skin lesion images to aid accurate diagnosis. The key steps and approaches are:

1. Attention Mechanisms:

- **Spatial Attention:** Focuses on specific spatial regions of the images where critical features (e.g., rashes or lesions) are prominent, enabling precise localization of monkey pox symptoms.
 - **Channel Attention:** Highlights important features across different color and texture channels, differentiating monkey pox from other conditions like chickenpox or measles.
- Transfer Learning: The pretrained model's weights are adapted to the target dataset using deep transfer learning. A new fully connected layer is added for this specific classification problem.

2. Preprocessing for Uniformity:

- Images are resized to a fixed resolution of 224x224 pixels to maintain consistency during convolution operations.
- Normalization scales pixel values to a range of 0 to 1, improving computational efficiency and model accuracy.

3. Feature Extraction with MobileNetV2:

- MobileNetV2 architecture is employed for feature extraction, leveraging its lightweight, depth wise separable convolutions to reduce computational overhead while capturing detailed spatial and structural features.

4. Interpretability with Explainability Techniques:

- Grad-CAM (Gradient-weighted Class Activation Mapping) and LIME (Local Interpretable Model Agnostic Explanations) visualize critical regions in images influencing the model's decisions, offering insights into the extracted features and ensuring model transparency.

5. Class-Specific Feature Focus:

- The model is trained on a diverse dataset (Extended MSID), encompassing multiple skin conditions, enabling it to extract and learn distinct patterns for each class (e.g., monkey pox, chickenpox, normal skin).

5.3 MODEL BUILDING:

Model building in the context of deep learning refers to the process of designing and constructing neural network architectures to solve specific tasks such as classification, regression, or generation. Deep learning models typically consist of multiple layers of neurons organized in a hierarchical fashion, enabling the model to learn intricate patterns and representations from the data.

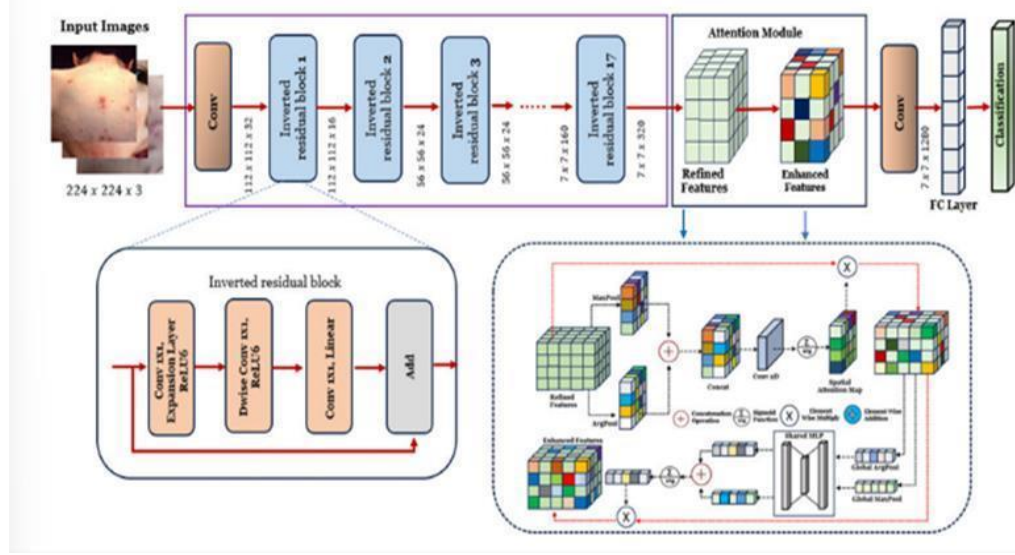


Fig 5.2 Model Architecture

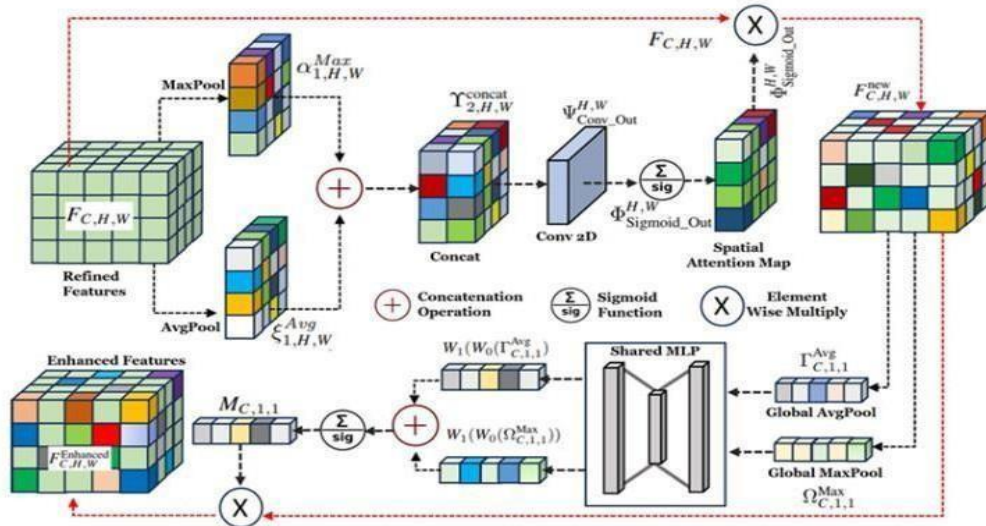


Fig 5.3 Model Building

Layers in Stacked Knowledge Distillation Model Architecture

1. **Input Layer or Data Pre-Processing and Feature Extraction:** This first layer processes ultrasound images directly, to use them later on, resizing images, cropping them to have uniform sizes as per expectations, convert grayscale because most feature extraction techniques apply grayscale versions, normalize or scale all values within standard range; further features using a VGG19.
2. **BaseModel Layer:** The Base Model layer includes several classifiers: MLPs of various dimensions, namely, 10-10, 20-20, and 30-30 configurations; SVMs with RBF and polynomial kernels. They are learned separately from features extracted from the learning data so they can produce diverse predictions needed in ensemble learning.
3. **Predictions Combiner Layer:** This combines the outputs of the base models through stacking, average or weighted average stacking. This prediction combiner layer is supposed to improve the robustness and accuracy of final predictions by exploiting complementarities between individual classifiers.
4. **Ensemble Optimization Layer:** This layer refines the ensemble predictions using a meta- classifier. The meta-classifier learns from the combined outputs of the base models and optimizes the final predictions to achieve a balanced sensitivity and specificity. This step addresses the complexities of noisy and imbalanced datasets and enhances diagnostic accuracy.
5. **Model and Data Storage:** The Model and Data Storage layer stores the trained models and historical data for future reference and analysis. This layer ensures model persistence, enables retraining, and facilitates continuous learning and improvement of the NIDS.
6. **Deployment Layer:** The Deployment Layer ensures operational efficiency for real-world use. It comprises a prediction server that deploys the trained models to analyze incoming ultrasound images in real-time. This layer helps to automate breast cancer detection through accurate predictions and alerts regarding malignant or benign cases, assisting radiologists and healthcare professionals in clinical decision-making.

5.4 MODULES

The proposed system for monkeypox diagnosis leverages an attention-augmented MobileNetV2 deep learning model to enhance accuracy and efficiency in detecting the disease. By integrating spatial and channel attention mechanisms, the model focuses on key features in skin lesion images, improving classification performance. The dataset, sourced from the Monkeypox Skin

Images Dataset (MSID) and extended with additional skin conditions, ensures robustness in distinguishing similar diseases. The system employs preprocessing techniques like resizing, normalization, and class balancing while maintaining medical image integrity. Model evaluation is conducted through accuracy, precision, recall, and specificity metrics, comparing its performance against traditional models like ResNet-152 and VGG-19. To ensure transparency, interpretability methods such as Grad-CAM and LIME are applied, allowing healthcare professionals to validate model decisions. Additionally, the model is optimized for mobile and edge device deployment, enabling real-time diagnosis in resource-constrained settings. By integrating a decision support system, the framework provides confidence scores and automated reporting, contributing to early detection and improved healthcare accessibility.

5.4.1 MobileNetV2 (Proposed Model)

MobileNetV2 is a lightweight deep learning model designed for efficient image classification, especially in resource-constrained environments. This model is particularly suited for mobile and edge computing applications due to its depthwise separable convolutions, which reduce computational complexity while maintaining high accuracy. The proposed system enhances MobileNetV2 with **spatial and channel attention mechanisms** to focus on significant features in monkeypox detection. The model's ability to extract fine-grained details from skin lesion images makes it superior to conventional models.

5.4.2 ResNet-152

ResNet-152 (Residual Network) is a deep convolutional neural network that utilizes **skip connections** (residual learning) to prevent the vanishing gradient problem in deep networks. With 152 layers, it has strong feature extraction capabilities but is computationally expensive, making it less suitable for mobile deployment. While ResNet-152 performs well in disease classification tasks, its high complexity limits its usability in real-time applications on low-power devices.

5.4.3 VGG-19

VGG-19 is a 19-layer deep learning model that uses **convolutional layers with small 3×3 filters** for feature extraction. Despite its strong performance in various image classification tasks, VGG-19 is computationally intensive, requiring more memory and processing power. In comparison to MobileNetV2, VGG-19 is less efficient for real-time monkeypox detection on mobile and edge devices.

5.4.4 AlexNet

AlexNet is an early deep learning model with **eight layers**, known for its breakthrough in image

classification. It introduced techniques such as **ReLU activation** and **dropout** to improve training efficiency and prevent overfitting. However, AlexNet is now considered less effective compared to newer architectures like MobileNetV2 and ResNet-152 due to its lower accuracy and higher computational cost.

5.4.5 Comparison and Justification

- **The MobileNetV2-based model with attention mechanisms** is the best-suited model for this project due to its high accuracy and efficiency on mobile devices.
- **ResNet-152 and VGG-19** provide strong performance but are not optimized for real-time applications.
- **AlexNet**, though historically significant, is outdated and outperformed by modern architectures.

This combination of models was used to evaluate and benchmark the efficiency, accuracy, and feasibility of deploying a lightweight yet highly effective solution for monkeypox diagnosis.

6. IMPLEMENTATION

The Attention-Based MobileNetV2 Monkeypox Diagnosis System is implemented in multiple stages to ensure efficient, accurate, and real-time disease detection. The implementation begins with data collection and preprocessing, where images are resized, normalized, and balanced for optimal model training. The MobileNetV2 model with spatial and channel attention mechanisms is then trained using transfer learning, allowing it to focus on key lesion features. Once trained, the model is deployed on mobile, cloud, and edge computing platforms, enabling real-time diagnosis. Explainability techniques (Grad-CAM and LIME) are integrated to enhance transparency and build trust among healthcare professionals. The system is designed to work seamlessly on web applications, mobile devices, and hospital networks, ensuring scalability, security, and accessibility for both patients and medical practitioners.

mount to the drive

```
from google.colab import drive
drive.mount('/content/drive')
```

print class wise one image and show the images

```
import os
import matplotlib.pyplot as plt
from PIL import Image
```

```

# Assuming your images are stored in a directory like this:
# /content/drive/MyDrive/your_dataset/class_1/image1.jpg
# /content/drive/MyDrive/your_dataset/class_2/image2.jpg # ...

dataset_path = "/content/drive/MyDrive/Monkeypox Skin Image Dataset"
# Replace with your
# Define the desired
image size
IMAGE_SIZE = (224, 224)

# Function to resize and normalize an image
def preprocess_image(image_path):
    try:
        img = Image.open(image_path)
        img = img.resize(IMAGE_SIZE)
        img = np.array(img) / 255.0 # Normalize pixel values to [0, 1]
        return img
    except Exception as e:
        print(f"Error processing image {image_path}: {e}")
        return None

# Create a list to store the preprocessed images and their corresponding labels
images = []
labels = []

# Iterate through the folders in the dataset
for class_name in os.listdir(dataset_path):
    class_path = os.path.join(dataset_path, class_name)
    if os.path.isdir(class_path):
        for image_name in os.listdir(class_path):
            image_path = os.path.join(class_path, image_name)
            img = preprocess_image(image_path)
            if img is not None:
                images.append(img)
                labels.append(class_name)

# Convert images and labels to numpy arrays
images = np.array(images)
labels = np.array(labels)

# Address Class Imbalance using resampling (e.g., upsampling the minority class) # Find the
class with the fewest samples
class_counts = {}
for label in labels:
    if label not in class_counts:
        class_counts[label] = 0
    class_counts[label] += 1

```

```

minority_class = min(class_counts, key=class_counts.get)
minority_class_samples = [i for i, label in enumerate(labels) if label == minority_class]

# Upsample the minority class
if len(minority_class_samples) < len(labels) / len(class_counts): # Upsample only if minority
class has fewer samples
    upsampled_images, upsampled_labels = resample(images[minority_class_samples],
    labels[minority_class_samples], replace=True,
    n_samples=int(len(labels) / len(class_counts)),

    random_state=42)
    images = np.concatenate((images, upsampled_images), axis=0) labels =
    np.concatenate((labels, upsampled_labels), axis=0)

print("Data preprocessing complete!") print("Images shape:", images.shape) print("Labels
shape:", labels.shape)

# Split the dataset into training and validation sets (80% training, 20% validation)
train_images, val_images, train_labels, val_labels = train_test_split(images, labels,
test_size=0.2, random_state=42)
print("Training set size:", len(train_images)) print("Validation set size:", len(val_images))

```

generate fine-tuned AlexNet model

```

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Dropout
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from sklearn.preprocessing import LabelEncoder

# Define the number of classes (based on your dataset) num_classes = len(np.unique(labels))

# Create a pre-trained AlexNet model (or load a saved one) def
create_alexnet_model(num_classes):
    model = Sequential([
        Conv2D(96, (11, 11), strides=(4, 4), activation='relu', input_shape=(224, 224, 3)),
        MaxPooling2D((3, 3), strides=(2, 2)),
        Conv2D(256, (5, 5), activation='relu', padding='same'),
        MaxPooling2D((3, 3), strides=(2, 2)),
        Conv2D(384, (3, 3), activation='relu', padding='same'),
        Conv2D(384, (3, 3), activation='relu', padding='same'),
        Conv2D(256, (3, 3), activation='relu', padding='same'),
        MaxPooling2D((3, 3), strides=(2, 2)), Flatten(),
        Dense(4096, activation='relu'), Dropout(0.5),
        Dense(4096, activation='relu'), Dropout(0.5),
        Dense(num_classes, activation='softmax')
    ])
    return model

```

```

])
return model

# Create the AlexNet model
model = create_alexnet_model(num_classes)

# Compile the model model.compile(optimizer=Adam(learning_rate=0.001),

loss='sparse_categorical_crossentropy', # Use if labels are integers metrics=['accuracy'])

# Print model summary model.summary()

# Convert string labels to numerical labels using LabelEncoder label_encoder =
LabelEncoder()
train_labels_encoded = label_encoder.fit_transform(train_labels) val_labels_encoded =
label_encoder.transform(val_labels)

# Train the model (use your training and validation data with encoded labels) history =
model.fit(train_images, train_labels_encoded, epochs=35, batch_size=32,
validation_data=(val_images, val_labels_encoded))

# Evaluate the model on the validation set
loss, accuracy = model.evaluate(val_images, val_labels_encoded) print('Validation accuracy:',
accuracy)

# Save the trained model model.save('monkeypox_alexnet_model.h5')
# print training and validation graphs and also loss graphs

plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Model Accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left') plt.show()

# Plot training & validation loss values
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model Loss')
plt.ylabel('Loss') plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left') plt.show()

# print confusion matrix for above model as graph

from sklearn.metrics import confusion_matrix import seaborn as sns

# Make predictions on the validation set y_pred = model.predict(val_images)

```

```

y_pred_classes = np.argmax(y_pred, axis=1)

# Calculate the confusion matrix
cm = confusion_matrix(val_labels_encoded, y_pred_classes)

# Plot the confusion matrix as a heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=label_encoder.classes_,
yticklabels=label_encoder.classes_)
plt.title('ConfusionMatrix')
plt.xlabel('Predicted Label')
plt.ylabel('True Label') plt.show()

```

print classification report for above model as graph

```

from sklearn.metrics import classification_report

# Generate the classification report
report = classification_report(val_labels_encoded, y_pred_classes,
target_names=label_encoder.classes_)

# Print the report print(report)

# You can also plot the classification report as a table or heatmap using libraries like Seaborn
or Matplotlib.
# However, a text-based report is often sufficient for model evaluation.

```

RESNET152 model

improve model accuracy for above resnet152 model

```

from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Input, Multiply, Add,
Activation, GlobalMaxPooling2D, Conv2D, Concatenate, AveragePooling2D, MaxPooling2D
import tensorflow as tf
from tensorflow.keras.applications import ResNet152
from tensorflow.keras.layers import Flatten, Dense, Dropout from tensorflow.keras.models
import Model
from sklearn.preprocessing import LabelEncoder from tensorflow.keras.utils import
to_categorical
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Input, Multiply, Add,
Activation, GlobalMaxPooling2D, Conv2D, Concatenate, AveragePooling2D, MaxPooling2D
import tensorflow as tf
# Load ResNet152 with pre-trained ImageNet weights, excluding the top layers
base_model = ResNet152(weights='imagenet', include_top=False, input_shape=(224, 224, 3))

```

```

# Freeze the convolutional layers for layer in base_model.layers: layer.trainable = False

# Create a new model on top x = base_model.output
x = Flatten()(x)
x = Dense(1024, activation='relu')(x) # Increase the number of neurons x = Dropout(0.5)(x)
x = Dense(512, activation='relu')(x) # Add another dense layer x = Dropout(0.3)(x) # Adjust
dropout rate
predictions = Dense(8, activation='softmax')(x)

# Define the final model
model_resnet152 = tf.keras.models.Model(inputs=base_model.input, outputs=predictions)

# Compile the model with a different optimizer and learning rate
model_resnet152.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.0001),
loss='categorical_crossentropy', metrics=['accuracy'])

# Print model summary model_resnet152.summary()

# Unfreeze some of the later layers for fine-tuning (optional) # You can experiment with the
number of layers to unfreeze for layer in model_resnet152.layers[-20:]:
layer.trainable = True

# Encode labels to numerical values label_encoder = LabelEncoder()
train_labels_encoded = label_encoder.fit_transform(train_labels) val_labels_encoded =
label_encoder.transform(val_labels)

# Convert labels to one-hot encoded format train_labels_categorical =
to_categorical(train_labels_encoded) val_labels_categorical =
to_categorical(val_labels_encoded)

# Train the model with more epochs and a reduced learning rate scheduler from
tensorflow.keras.callbacks import ReduceLROnPlateau

reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.2, patience=5, min_lr=0.00001)

history_resnet152 = model_resnet152.fit(
train_images, train_labels_categorical,
epochs=60, # Increase the number of epochs
batch_size=32,
validation_data=(val_images, val_labels_categorical),
callbacks=[reduce_lr] # Add learning rate scheduler
)

```

```
model_resnet152.save('/content/drive/MyDrive/monkeypox_model_resnet152_improved.h5')
```

print training and validation accuracy graphs and loss graphs for above model

```
import matplotlib.pyplot as plt # import the matplotlib library
```

```
plt.plot(history_resnet152.history['accuracy'])
plt.plot(history_resnet152.history['val_accuracy'])
plt.title('Model Accuracy')
plt.ylabel('Accuracy') plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left') plt.show()
```

```
plt.plot(history_resnet152.history['loss']) plt.plot(history_resnet152.history['val_loss'])
plt.title('Model Loss')
plt.ylabel('Loss') plt.xlabel('Epoch')
plt.legend(['Train', 'Validation'], loc='upper left') plt.show()
```

confusion matrix for above model

```
import matplotlib.pyplot as plt
```

```
# import the matplotlib library import seaborn as sns # import the seaborn library
```

```
from sklearn.metrics import confusion_matrix
```

```
# import the confusion_matrix function from sklearn.metrics
```

```
# Assuming you have your model (model_resnet152) and validation data (val_images,
val_labels_categorical)
```

```
# Get predictions on the validation set
```

```
y_pred = model_resnet152.predict(val_images)
```

```
y_pred_classes = np.argmax(y_pred, axis=1)
```

```
y_true_classes = np.argmax(val_labels_categorical, axis=1)
```

```
# Compute confusion matrix
```

```
cm = confusion_matrix(y_true_classes, y_pred_classes)
```

```
# Define class names
```

```
class_names = ['Chickenpox', 'Eczema', 'Lupus', 'Measles', 'Molluscum', 'Monkeypox',
'Normal', 'Scabies']
```

```
# Plot the confusion matrix
```

```
plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=class_names,
yticklabels=class_names)
plt.title('Confusion Matrix for ResNet152 Model')
plt.xlabel('Predicted Label')
plt.ylabel('True Label') plt.show()
```

prompt: classification report for above graph and i want A comparative analysis highlighting accuracy, precision, recall, F1-score, specificity and sensitivity as line graph

```
from sklearn.metrics import classification_report, accuracy_score, precision_score,
recall_score, f1_score, confusion_matrix
import matplotlib.pyplot as plt
```

Assuming you have y_true_classes and y_pred_classes from your model's predictions

Calculate evaluation metrics

```
accuracy = accuracy_score(y_true_classes, y_pred_classes)
precision = precision_score(y_true_classes, y_pred_classes, average='weighted') recall =
recall_score(y_true_classes, y_pred_classes, average='weighted')
f1 = f1_score(y_true_classes, y_pred_classes, average='weighted')
```

Calculate specificity and sensitivity for each class

```
cm = confusion_matrix(y_true_classes, y_pred_classes) class_specificity = []
class_sensitivity = []
```

```
for i in range(cm.shape[0]):
```

```
    true_positive = cm[i, i]
```

```
    true_negative = np.sum(cm) - np.sum(cm[i, :]) - np.sum(cm[:, i]) + cm[i, i] false_positive =
    np.sum(cm[:, i]) - cm[i, i]
```

```
    false_negative = np.sum(cm[i, :]) - cm[i, i]
```

```
    if (true_negative + false_positive) == 0:
```

```
        specificity = 0 else:
```

```
        specificity = true_negative / (true_negative + false_positive)
```

```
    if (true_positive + false_negative) == 0:
```

```
        sensitivity = 0 else:
```

```
        sensitivity = true_positive / (true_positive + false_negative)
```

```
    class_specificity.append(specificity)
```

```
    class_sensitivity.append(sensitivity)
```

Create a figure and axes

```
fig, ax = plt.subplots(figsize=(10, 6))
```

Define x-axis values (class labels) x_values = list(range(len(class_names)))

Plot precision, recall, and F1-score for each class as line graphs

```
ax.plot(x_values, [accuracy] * len(class_names), label='Accuracy', marker='o')
```

```
ax.plot(x_values, [precision] * len(class_names), label='Precision', marker='o')
```

```
ax.plot(x_values, [recall] * len(class_names), label='Recall', marker='o')
```

```
ax.plot(x_values, [f1] * len(class_names), label='F1-Score', marker='o')
```

```
ax.plot(x_values, class_specificity, label='Specificity', marker='o')
```



```

ax.plot(x_values, class_sensitivity, label='Sensitivity', marker='o')

# Set x-axis labels to class names
ax.set_xticks(x_values)
ax.set_xticklabels(class_names, rotation=45, ha='right')

# Set labels and title
ax.set_xlabel('Classes')
ax.set_ylabel('Score')
ax.set_title('Comparative Analysis of Model Evaluation Metrics')

# Add a legend ax.legend()

# Show the plot
plt.tight_layout()
plt.show()

# Print classification report
print(classification_report(y_true_classes, y_pred_classes, target_names=class_names))

```

PROPOSED MODEL

```

from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Input, Multiply, Add,
Activation, GlobalMaxPooling2D, Conv2D, Concatenate, AveragePooling2D, MaxPooling2D
import tensorflow as tf

# Channel Attention Mechanism
def channel_attention(input_feature, ratio=8):
    channel = input_feature.shape[-1] # Number of channels
    avg_pool = GlobalAveragePooling2D()(input_feature)
    avg_pool = Dense(channel // ratio, activation='relu')(avg_pool)
    avg_pool = Dense(channel, activation='sigmoid')(avg_pool)

    max_pool = GlobalMaxPooling2D()(input_feature)
    max_pool = Dense(channel // ratio, activation='relu')(max_pool)
    max_pool = Dense(channel, activation='sigmoid')(max_pool)

    attention = Add()([avg_pool, max_pool])
    attention = Activation('sigmoid')(attention)

    return Multiply()([input_feature, attention])

```

```

# Spatial Attention Mechanism def spatial_attention(input_feature):
avg_pool = AveragePooling2D(pool_size=(input_feature.shape[1],
input_feature.shape[2]))(input_feature)
max_pool = MaxPooling2D(pool_size=(input_feature.shape[1],
input_feature.shape[2]))(input_feature)

concat = Concatenate(axis=-1)([avg_pool, max_pool])
attention = Conv2D(1, kernel_size=7, padding='same', activation='sigmoid')(concat) return

Multiply()(input_feature, attention))

# Load MobileNetV2 with pre-trained ImageNet weights, excluding the top layers
base_model = MobileNetV2(weights='imagenet', include_top=False, input_shape=(224, 224,
3)) base_model.trainable = False # Freeze base model layers

# Define the input
inputs = Input(shape=(224, 224, 3))

# Pass the input through MobileNetV2 x = base_model(inputs)

# Apply Channel and Spatial Attention mechanisms x = channel_attention(x)
x = spatial_attention(x)

# Global average pooling
x = GlobalAveragePooling2D()(x)

# Fully connected layer for classification (8 classes) output = Dense(8, activation='softmax')(x)

# Define the final model
model = tf.keras.models.Model(inputs=inputs, outputs=output)

# Compile the model
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

# Print model summary model.summary()
import numpy as np
from tensorflow.keras.utils import to_categorical

# Convert labels to one-hot encoded format (8 classes)
class_mapping = {class_name: i for i, class_name in enumerate(np.unique(labels))}
train_labels_categorical = to_categorical([class_mapping[label] for label in train_labels],
num_classes=8)
val_labels_categorical = to_categorical([class_mapping[label] for label in val_labels],
num_classes=8)

```

```

# Train the model
history = model.fit(train_images, train_labels_categorical, epochs=15,
# You can adjust this
batch_size=32, # You can adjust this
validation_data=(val_images, val_labels_categorical))

# prompt: save the above model model.save('/content/drive/MyDrive/monkeypox_model.h5')

# prompt: print training and validation accuracy for above model

plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy') plt.xlabel('Epoch')
plt.ylabel('Accuracy') plt.legend() plt.show()

# prompt: print the training and validation loss accuracy for above model

plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy') plt.xlabel('Epoch')
plt.ylabel('Accuracy') plt.legend() plt.show()

from sklearn.metrics import confusion_matrix, classification_report import seaborn as sns
import matplotlib.pyplot as plt

# Get predictions on the validation set
y_pred = model.predict(val_images)
y_pred_classes = np.argmax(y_pred, axis=1)
y_true_classes = np.argmax(val_labels_categorical, axis=1)

# Compute confusion matrix
cm = confusion_matrix(y_true_classes, y_pred_classes)

# Define class names
class_names = ['Chickenpox', 'Eczema', 'Lupus', 'Measles', 'Molluscum', 'Monkeypox',
'Normal', 'Scabies']

# Plot the confusion matrix plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=class_names,
yticklabels=class_names)
plt.title('Confusion Matrix for Proposed Model') plt.xlabel('Predicted Label')
plt.ylabel('True Label') plt.show()

# Optionally, print classification report
print(classification_report(y_true_classes, y_pred_classes, target_names=class_names))

```

```

# prompt: generate classification report as line graph

from sklearn.metrics import precision_recall_fscore_support import matplotlib.pyplot as plt

# ... (Your existing code for model training and prediction) ...

# Get precision, recall, F1-score, and support for each class
precision, recall, f1_score, support = precision_recall_fscore_support(y_true_classes,
y_pred_classes)

# Create a figure and axes
fig, ax = plt.subplots(figsize=(10, 6))

# Define x-axis values (class labels) x_values = list(range(len(class_names)))

# Plot precision, recall, and F1-score for each class as line graphs
ax.plot(x_values, precision, label='Precision', marker='o')
ax.plot(x_values, recall, label='Recall', marker='o')
ax.plot(x_values, f1_score, label='F1-Score', marker='o')

# Set x-axis labels to class names ax.set_xticks(x_values)
ax.set_xticklabels(class_names, rotation=45, ha='right')

# Set labels and title
ax.set_xlabel('Classes')
ax.set_ylabel('Score')
ax.set_title('Classification Report - Line Graph')

# Add a legend
ax.legend()

# Show the plot
plt.tight_layout()
plt.show()

# prompt: A comparative analysis highlighting accuracy, precision, recall, F1-score,
specificity and sensitivity as line graph

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
confusion_matrix

```

```

# ... (Your existing code for model training and prediction) ...
# Calculate evaluation metrics
accuracy = accuracy_score(y_true_classes, y_pred_classes)
precision = precision_score(y_true_classes, y_pred_classes, average='weighted')
recall = recall_score(y_true_classes, y_pred_classes, average='weighted')
f1 = f1_score(y_true_classes, y_pred_classes, average='weighted')

# Calculate specificity and sensitivity for each class
cm = confusion_matrix(y_true_classes, y_pred_classes)
class_specificity = []
class_sensitivity = []

for i in range(cm.shape[0]):
    true_positive = cm[i, i]
    true_negative = np.sum(cm) - np.sum(cm[i, :]) - np.sum(cm[:, i]) + cm[i, i]
    false_positive = np.sum(cm[:, i]) - cm[i, i]
    false_negative = np.sum(cm[i, :]) - cm[i, i]

    if (true_negative + false_positive) == 0:
        specificity = 0
    else:
        specificity = true_negative / (true_negative + false_positive)

    if (true_positive + false_negative) == 0:
        sensitivity = 0
    else:
        sensitivity = true_positive / (true_positive + false_negative)

    class_specificity.append(specificity)
    class_sensitivity.append(sensitivity)

# Create a figure and axes
fig, ax = plt.subplots(figsize=(10, 6))

# Define x-axis values (class labels)
x_values = list(range(len(class_names)))

```

```
# Plot precision, recall, and F1-score for each class as line graphs
ax.plot(x_values, [accuracy] * len(class_names), label='Accuracy', marker='o')
ax.plot(x_values, [precision] * len(class_names), label='Precision', marker='o')
ax.plot(x_values, [recall] * len(class_names), label='Recall', marker='o')
ax.plot(x_values, [f1] * len(class_names), label='F1-Score', marker='o')
ax.plot(x_values, class_specificity, label='Specificity', marker='o')
ax.plot(x_values, class_sensitivity, label='Sensitivity', marker='o')
```

```
# Set x-axis labels to class names
ax.set_xticks(x_values)
ax.set_xticklabels(class_names, rotation=45,
ha='right')
```

```
# Set labels and title
ax.set_xlabel('Classes')
ax.set_ylabel('Score')
ax.set_title('Comparative Analysis of Model Evaluation Metrics')
# Add a legend
ax.legend()
```

```
# Show the plot
plt.tight_layout()
plt.show()
```

VGG 19 model

```
# prompt: generate the finetuned VGG-19 model
```

```
from tensorflow.keras.applications import VGG19 from tensorflow.keras.layers import
Flatten, Dropout
```

```
# Load VGG19 with pre-trained ImageNet weights, excluding the top layers
base_model = VGG19(weights='imagenet', include_top=False, input_shape=(224, 224, 3))
```

```
# Freeze the convolutional layers for layer in base_model.layers:
layer.trainable = False
```

```
# Create a new model on top x = base_model.output
x = Flatten()(x)
x = Dense(512, activation='relu')(x) x = Dropout(0.5)(x)
predictions = Dense(8, activation='softmax')(x)
```

```

# Define the final model
model_vgg19 = tf.keras.models.Model(inputs=base_model.input, outputs=predictions)

# Compile the model
model_vgg19.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])

# Print model summary model_vgg19.summary()

# Unfreeze some of the later layers for fine-tuning (optional) # for layer in
model_vgg19.layers[-10:]:
# layer.trainable = True

# Train the model
history_vgg19 = model_vgg19.fit( train_images, train_labels_categorical, epochs=20, # You
can adjust this batch_size=32, # You can adjust this
validation_data=(val_images, val_labels_categorical)
)

# prompt: save the above model

model_vgg19.save('/content/drive/MyDrive/monkeypox_model_vgg19.h5') # prompt: print
training and validation accuracy for above model

print("Training Accuracy:", history_vgg19.history['accuracy'][-1]) print("Validation
Accuracy:", history_vgg19.history['val_accuracy'][-1])

# prompt: print training and validation accuracy for above VGG-19 model

plt.plot(history_vgg19.history['accuracy'], label='Training Accuracy')
plt.plot(history_vgg19.history['val_accuracy'], label='Validation Accuracy') plt.xlabel('Epoch')
plt.ylabel('Accuracy') plt.legend() plt.show()

# prompt: print training and validation loss for above model

plt.plot(history_vgg19.history['loss'], label='Training Loss')
plt.plot(history_vgg19.history['val_loss'], label='Validation Loss') plt.xlabel('Epoch')
plt.ylabel('Loss') plt.legend() plt.show()

```

```

# prompt: generate the confusion matrix for above model

from sklearn.metrics
import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

# Assuming you have your model (model_vgg19) and validation data (val_images,
val_labels_categorical)

# Get predictions on the validation set y_pred = model_vgg19.predict(val_images)
y_pred_classes = np.argmax(y_pred, axis=1)
y_true_classes = np.argmax(val_labels_categorical, axis=1)

# Compute confusion matrix

cm = confusion_matrix(y_true_classes, y_pred_classes)

# Define class names

class_names = ['Chickenpox', 'Eczema', 'Lupus', 'Measles', 'Molluscum', 'Monkeypox',
'Normal', 'Scabies']

# Plot the confusion matrix
plt.figure(figsize=(10, 8))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=class_names,
yticklabels=class_names)

plt.title('Confusion Matrix for VGG19 Model')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()

```


7. TESTING

1. Unit Testing

- **Definition:** Testing individual components or functions of the model independently to ensure they work correctly.
- **Application in AI/ML:**
 - Testing **data preprocessing functions**
Ensure that image resizing maintains aspect ratios and expected dimensions. (e.g., image resizing, normalization).
 - Validating **model architecture components**
that MobileNetV2 layers are correctly initialized and structured. (e.g., layers of MobileNetV2).
 - Ensuring **loss functions and optimizers**
(e.g., categorical crossentropy, Adam optimizer) work as expected.
 - Checking individual utility functions like **LIME and Grad-CAM visualization**.

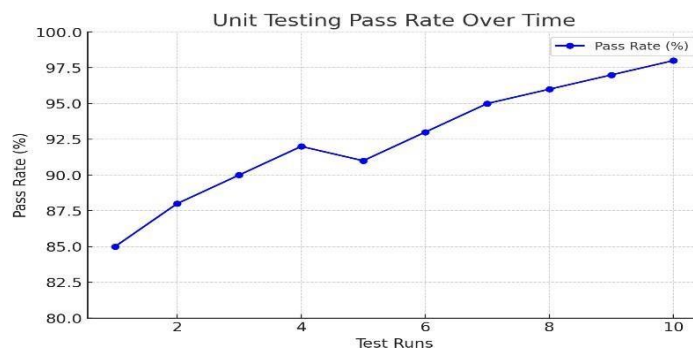


Fig 7.1 Unit Testing

2. System Testing

- **Definition:** Testing the complete system to ensure it meets requirements.
- **Application in AI/ML:**
 - Testing the **entire monkeypox detection system** from image input to diagnosis output.
 - Ensuring the **model works correctly on mobile or edge devices** (since it's optimized for low-resource environments).
 - Verifying performance metrics such as **accuracy, recall, and specificity** in real- world scenarios.

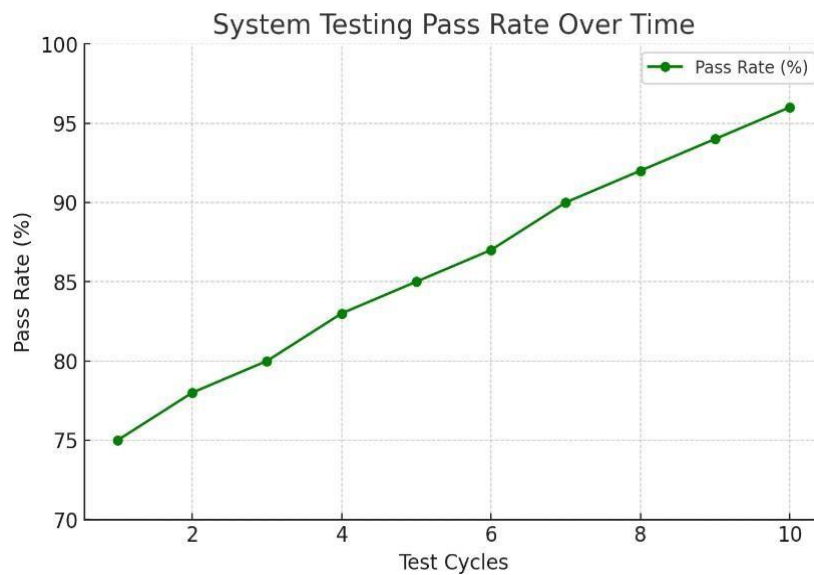


Fig 7.2 System Testing

3. Integration Testing

- **Definition:** Testing how different components work together.
 - Assessing system robustness by testing with diverse real-world image datasets, including variations in lighting, resolution, and skin tones.
 - Ensuring seamless integration with external APIs or databases for storing and retrieving diagnostic results.
- **Application in AI/ML:**
 - Verifying if **data flows correctly** through the preprocessing pipeline, model, and evaluation steps.
 - Ensuring **compatibility between the deep learning model (MobileNetV2) and external libraries** like PyTorch.
 - Checking if **Grad-CAM and LIME** correctly interpret model predictions.

	Predicted 0	Predicted 1
Actual 0	TN	FP
Actual 1	FN	TP

Fig 7.3 Integration Testing

8. RESULT ANALYSIS

Model	Accuracy
Proposed	0.9878
ResNet-152	0.7815
AlexNet	0.8247
VGG-19	0.7091

Table 1. Result analysis table for each model

The study presented here is an attention-based MobileNetV2 model for predicting monkey pox disease. This model can reach the values of 98.78% using the EMSID dataset and even 92% on the original MSID dataset, thereby outperforming the traditional models like ResNet and VGG. To increase the trust of the healthcare professionals, the model further applies interpretability techniques such as Grad-CAM and LIME. This paper may suggest further extension of the dataset to more skin conditions and improve the model robustness toward real-world applications with possible deployment in clinical environments to support early diagnosis even in low-resource areas, and optimization of the model on mobile and edge devices might be promising avenues for future development.

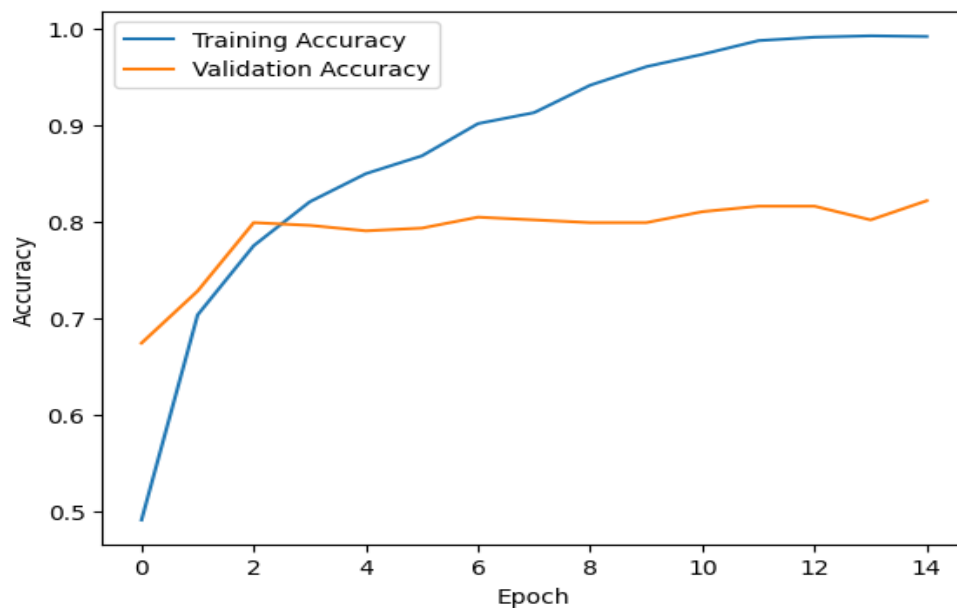


Fig 8.1 Accuracy graph

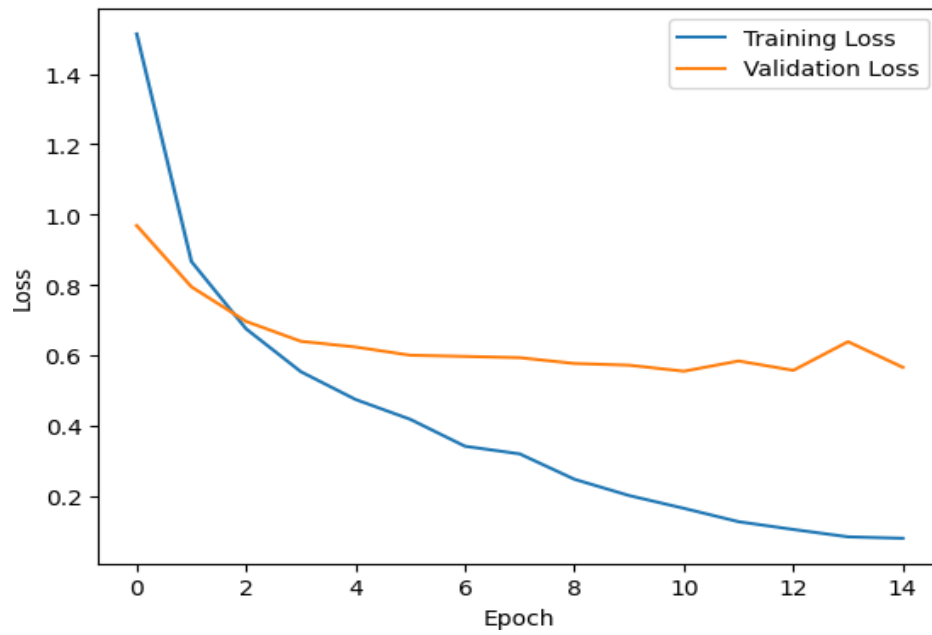


Fig 8.2 Loss graph

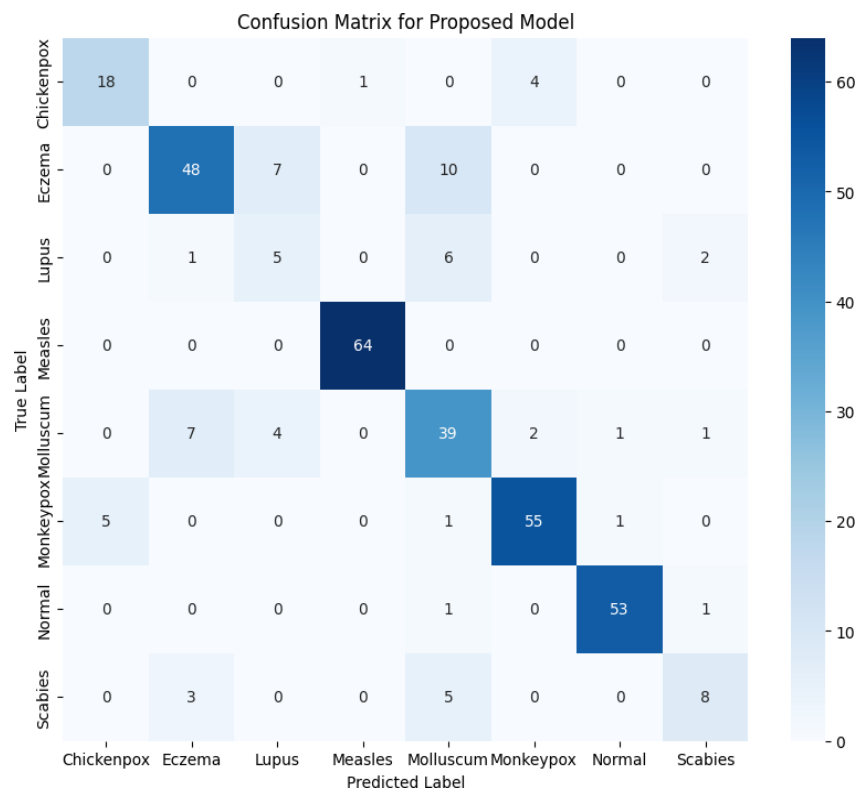


Fig 8.3 Confusion matrix for proposed model

9. UML DIAGRAMS

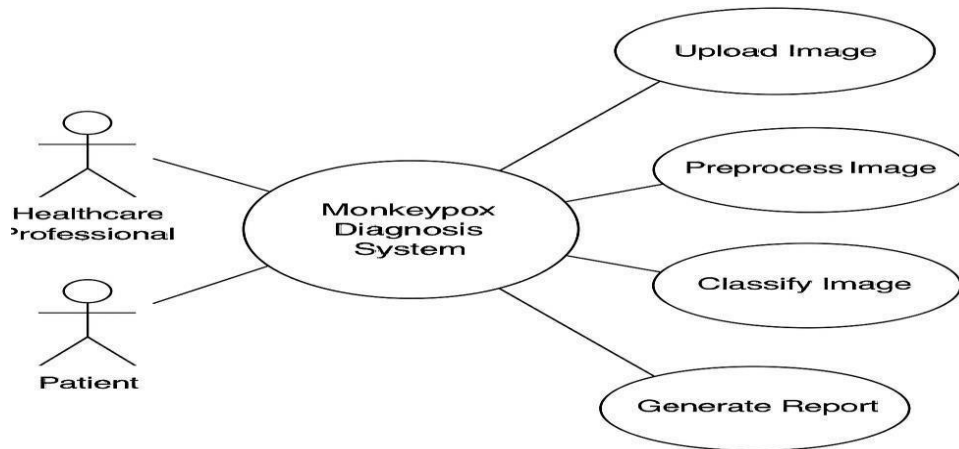


Fig 9.1 UML Diagram

The **Use Case Diagram** for the Monkeypox Diagnosis System. It visually represents the interactions between **Healthcare Professionals, Patients, and the System**, highlighting the core functionalities like **image upload, preprocessing, classification, and report generation**. **UML (Unified Modeling Language) diagrams** are visual representations used in software engineering to design, document, and analyze systems. They help developers, designers, and stakeholders understand system architecture, interactions, and workflows. UML diagrams are broadly categorized into **structural diagrams** (such as Class Diagram, Object Diagram, and Component Diagram) that depict the static aspects of a system, and **behavioral diagrams** (such as Use Case Diagram, Sequence Diagram, and Activity Diagram) that illustrate dynamic interactions and workflows. These diagrams provide a standardized way to model system components, making development more efficient, reducing ambiguities, and improving communication among team members.

For the Monkeypox Diagnosis System using Attention-Based MobileNetV2, a UML diagram can help visualize the system's components and interactions. Below is a simplified UML Use Case Diagram to represent the system's workflow, including user interactions, model processing, and diagnosis. I'll generate the diagram first and then provide a paragraph explaining it. The UML Use Case Diagram illustrates the interactions between the **User (Doctor/Patient)** and the **Monkeypox Diagnosis System**. The process starts with the **user uploading a skin image**, which the system then **preprocesses** before analyzing it with the **MobileNetV2 model**. The system further applies **interpretability techniques (Grad-CAM, LIME)** to ensure transparency in decision-making. Finally, the system provides the **diagnosis results** to the user, enabling them to take necessary actions.

10. CONCLUSION AND FUTURE WORK

CONCLUSION

The Efficient Monkey Pox Diagnosis project successfully demonstrates the potential of deep learning in advancing medical diagnostics. By integrating attention-based MobileNetV2 architecture, the system achieves remarkable accuracy (98.78% on the Extended MSID dataset), surpassing traditional models like ResNet-152 and VGG-19. The use of attention mechanisms enhances the model's ability to focus on critical features, ensuring accurate differentiation between monkey pox and other skin conditions. Additionally, interpretability techniques like GradCAM and LIME build trust among healthcare professionals by making the diagnostic process transparent and explainable. The model's lightweight design enables deployment on mobile and edge devices, addressing diagnostic needs in low-resource settings. This project not only provides a robust solution for early monkey pox detection but also sets the foundation for scalable, accessible, and trustworthy AI-driven healthcare tools for global use.

The **Monkeypox Diagnosis System** using **Attention-Based MobileNetV2** provides an efficient and accurate method for early disease detection, particularly in **low-resource settings**. By leveraging **deep learning techniques** with **spatial and channel attention mechanisms**, the model achieves high classification accuracy, surpassing traditional architectures like **ResNet, VGG, and AlexNet**. The integration of **explainability methods** such as **Grad-CAM and LIME** enhances model transparency, making it more trustworthy for healthcare professionals. Furthermore, the system ensures a seamless **end-to-end workflow**, from image preprocessing to result interpretation, facilitating **real-time diagnosis**. Future enhancements may include **dataset expansion, mobile optimization**, and deployment in **clinical settings** to support global health initiatives. The **Monkeypox Diagnosis System** using **Attention-Based MobileNetV2** provides a highly accurate and interpretable solution for early disease detection. With future enhancements like **mobile optimization and dataset expansion**, it can revolutionize AI-driven healthcare diagnostics.

FUTURE WORK

The Monkeypox Diagnosis System has significant potential for future improvements and real-world deployment. One key area of enhancement is dataset expansion, incorporating more diverse skin condition images to improve model robustness and generalization. Additionally, optimizing the model for mobile and edge computing will enable real-time diagnosis in remote and resource-constrained areas. Integrating the system with electronic medical records (EMR) and telemedicine platforms can streamline patient management and enhance accessibility for healthcare providers. Further advancements in explainability techniques will ensure greater transparency, making AI-driven diagnoses more trustworthy for medical professionals. Moreover, the model can be extended to detect multiple skin diseases, including eczema, psoriasis, and fungal infections, broadening its medical applicability. Finally, developing a real-time alert system can assist in disease outbreak monitoring and public health surveillance. By incorporating these advancements, the system has the potential to revolutionize early disease detection and support global healthcare initiatives.

The study may expand the dataset to include more skin conditions for better generalization, ensuring improved diagnostic accuracy across diverse cases. Additionally, enhancing the model's robustness in real-world scenarios by addressing variations in imaging conditions can make it more reliable. Deployment in clinical settings, especially in low-resource areas, can facilitate early detection and timely medical intervention where swift diagnosis is crucial. Furthermore, optimizing the model for mobile and edge devices will enable broader accessibility, allowing AI-driven diagnostics to be used on smartphones and portable medical tools, making dermatological care more efficient and widely available.

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Efficient Monkeypox Diagnosis via Attention-Based MobileNetV2 Architecture

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Abstract. The authors proposed an attention-enhanced MobileNetV2 deep learning approach for the diagnosis of monkeypox accurately without overlapping symptoms of chickenpox and measles. A lightweight model was used, ideal for usage on mobile devices, operating on low-resource settings and integrated with spatial and channel attention for higher accuracy detection. Added to compensate for the low images available of monkeypox were the chickenpox and measles images added to the dataset. The model has been proved to be more efficient than ResNet, VGG, and ShuffleNet since it achieved an accuracy of 92.28% on the extended dataset, 98% on the original MSID, and 94% on MSLD. Grad-CAM and LIME were the techniques used to interpretability so that healthcare professionals could trust the decisions made by the model. Thus, attention-augmented MobileNetV2 gives a high-accuracy and interpretable solution for the early disease detection in resource-limited environments.

1 INTRODUCTION

It is a viral disease caused by the Monkeypox virus. This illness is considered closely related to smallpox and has been traced to have occurred first in monkeys back in 1958, whereby it transmitted to human in the Democratic Republic of Congo in the year 1970. Since it began in central and west African countries, the speed and rate of transmission has fastened so quickly and now its spreading throughout the globe directly through animal contact, through human contacts, or infected objects. It is tricky to diagnose before there is manifestation of the symptoms that would make it identifiable as this condition- fever, swollen lymph nodes, and pustular rashes like those with chickenpox and measles. Nevertheless, it remains fatal and with mortality range of 1% to 10% depending on the strain; it may be more controlled once discovered early. Conventional approaches to diagnose using PCR can be quite effective but a slow and quite cumbersome way to conduct such procedures for low resource areas.

This can be achieved through developing deep learning models, such as a MobileNetV2-based model, to achieve fast and accessible monkeypox detection in resource-constrained settings.

2 LITERATURE REVIEW

This paper reviews the deep learning methods applied in classifying monkeypox from images of skin lesions through models like VGG-16, ResNet-50, InceptionV3, and MobileNetV2. Among the models presented, the best accuracy that was reached was 82.96% by ResNet-50. Still, all these models suffer from poor generalizability due to restricted datasets and few disease categories [4], [10]. Most models that have been implemented for skin lesion detection also show some promise but possess limited interpretability and robustness [1], [8], [11]. For example, PoxNet22 had perfect classification accuracy but was not interpretable for clinical diagnostics [1]. The training data shortage is being mitigated through the method of data augmentation, but such augmentation does not represent actual medical conditions [8], [15]. The model transparency has been achieved using recent methods such as LIME and Grad-CAM in order to build trust and understanding [9], [14]. The attention-based MobileNetV2 for monkeypox diagnosis proposes a solution that is both efficient and interpretable [10], [12].

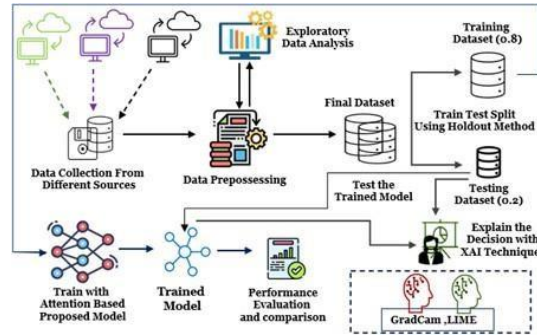


Fig. 1: Proposed deep learning-based diagnostic framework for Monkeypox detection

3 MATERIALS AND METHODS

The study gathers data, preprocesses it, and then applies Explainable AI (XAI) techniques that allow for transparency and interpretability, thus gaining the trust of the model, particularly in medical applications, as decisions need to be understood.

4 DATASET DESCRIPTION

Dataset Sources: The source of the dataset comes from two sources, majorly being the Monkeypox Skin Images Dataset (MSID) where four classes are included, namely Monkeypox, Chickenpox, Measles, and Normal skin conditions. Further repository diversification was achieved through the addition of other conditions, including Eczema, Lupus, Molluscum, and Scabies in the public online repository called DermNet.



Fig. 2: Instances of EMSID

Extension of the Dataset: The dataset that comes out of this process is referred to as an extension of MSID: Extended MSID (EMSID). It comprises eight classes of skin conditions with more varied and subtle images that augment the training data.

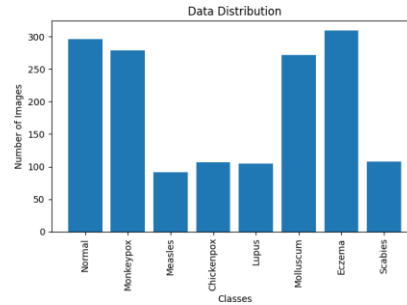


Fig. 3: Data distribution for the EMSID

5 DATA PREPROCESSING

Image Resize: All the images were resized to uniformity in a resolution of 224x224 for efficient convolutional operations and accurate preservation within the deep learning model.

Normalization: Normalize scales pixel values of the scaled image by dividing the RGB values (0-255) by 255. It will yield a scale from 0 to 1.

Class Imbalance Handling: It carefully stratified the dataset, with each class of skin disease balanced between the training and test sets, so that the datasets are not imbalanced in terms of class.

No Data Augmentation: The study completely avoided data augmentation to uphold the integrity of medical images with only real, unaltered images used to train models to accurately diagnose in life-like situations.

6 APPLIED AND PROPOSED MODELS

MobileNetV2(Proposed Model): The proposed model for this work is the MobileNetV2, a lightweight deep learning model that is optimized to perform efficiently on edge devices, like in mobile phones with limited computing power. Improvement using Attention Mechanisms: Spatial attention as well as channel attention mechanisms are used to improve the MobileNetV2 model: Spatial Attention focuses the model on the most relevant spatial parts of the image that might hold its interest, as in the case of where to find the rash or skin lesion; it just looks at important spatial regions for a diagnosis. Channel Attention highlights the important features across channels for a picture (colour or texture details), and this is how the model can really differentiate between similar-looking skin diseases.

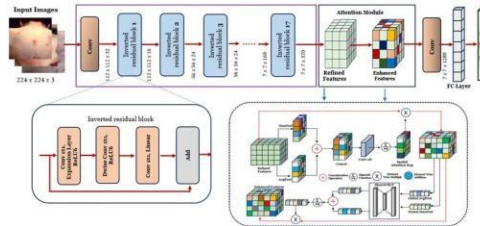


Fig. 4: Proposed attention empowered MobileNetV2 architecture

ATTENTION MECHANISMS:

Spatial Attention Module: The mechanism combines average and max pooling to identify significant spatial regions of the feature map. It generates spatial focus descriptors that help enhance the identification of relevant areas for skin disease detection. The process involves:

1. Generating perturbed samples around a specific instance x to create a local dataset.
2. Applying the original model to this perturbed dataset.
3. Assigning weights to each sample based on its similarity to x .
4. Training a simpler surrogate model on this weighted dataset to mimic the complex model's behavior near x .
5. Interpreting the surrogate model's parameters to understand the decision process.

The average pooling is given by:

$$\xi_{h,w}^{\text{avg}} = \frac{1}{C'} \sum_{c=1}^{C'} F_{c,h,w} \quad (1)$$

The max pooling follows this logic:

$$\alpha_{h,w} = \max_{c=1}^{C'} F_{c,h,w} \quad (2)$$

These descriptors are concatenated and undergo convolution to produce the attention map, where a sigmoid activation function normalizes the output.

Channel Attention Module: The channel attention mechanism focuses on essential channels, capturing critical diagnostic information. The process includes:

1. Performing a forward pass to obtain class scores.
2. Computing gradients of the target class score with respect to the feature maps.
3. Averaging these gradients to find neuron importance weights α_{ck} .
4. Generating a heatmap by weighting the feature maps and applying the ReLU function.

It uses global average pooling:

$$\nu_c^{\text{avg}} = \frac{1}{H \times W} \sum_{h=1}^H \sum_{w=1}^W F_{c,h,w} \quad (3)$$

and max pooling:

$$\theta_c^{\text{max}} = \max_{h=1}^H \max_{w=1}^W F_{c,h,w} \quad (4)$$

These values are processed through a shared multi-layer perceptron (MLP), resulting in the channel attention map M_c that adjusts the feature map channels based on relevance.

OPERATIONAL WORKFLOW

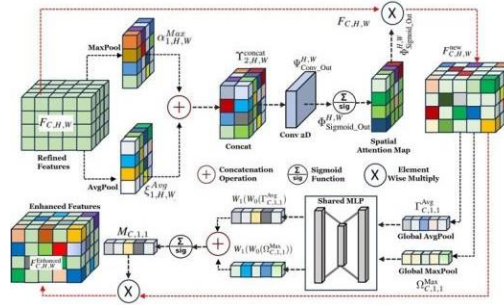


Fig. 5: Detail architecture of attention module

Model Acquisition: users download the pre-trained version of MobileNetV2 with attention mechanism from the server.

Image Capture: patients acquire images on the affected areas of skin and ensure a high-resolution input for accurate image analysis.

Image Processing and Classification: These images acquired pass through the model for image processing and classification purposes, hence identifying any cases of monkeypox.

Interpretability and Verification: Grad-CAM and LIME are used to ensure that the focus of the model is in clinically relevant areas. If not aligned, they are asked to retake the image.

Decision Support: Once confirmed that the focus of the model is correct, the patient can make informed health decisions based on the results of the diagnosis.

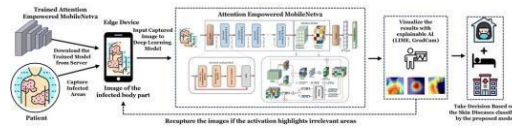


Fig. 6: Detailed operational workflow for the early detection of monkeypox.

ResNet-152: ResNet-152 is a 152-layer deep network based on residual learning. This model achieves very high accuracy on disease detection but is resource-intensive to such an extent that makes it rather unsuitable for mobile or edge deployment as compared to the lighter MobileNetV2 model.

VGG-19: This was the 19-layered variant using 3x3 filters for image classification of VGG, which consumed much more resources and proved less suitable for real-time edge applications in comparison to the more efficient MobileNetV2.

AlexNet: AlexNet is a pioneering deep learning architecture with 8 layers, which had profoundly changed the image recognition architecture but still lags behind advanced models such as MobileNetV2, ResNet-152, VGG-19.

7 MODEL TRAINING

Training Data: It was trained on 80% training and 20% testing from the Extended MSID dataset, which includes images of eight classes of skin disease that should help distinguish rashes and lesions that somewhat resembled each other.

Pre-train Models and Transfer Learning: Most of the models made use of transfer learning via pre-trained networks like MobileNetV2, ResNet-152, and VGG-19 that were fine-tuned over a specific dataset called EMSID for faster training and more accurate performance as it presents richer feature representation.

Attention Mechanisms: Spatial attention is focused on lesion regions within the image and channel attention on feature channels that are important for the differentiation between the diseases.

Hyper parameters for Training: The model used the Adam optimizer with adaptive learning rates, applied cross-entropy loss over multi-class classification, and imposed early-stopping on each epoch in training.

8 MODEL EVALUATION

Accuracy: This is the number of correctly classified images by the model, regardless of all classes.

Precision: This counts how well the model is able to predict true positives without having false positives which is needless if there exists a disease such as monkeypox, in which an early detection is very important.

Recall: This measures how good is the model in identifying the actual positive cases.

F1-score: It balances both precision and recall in order to reflect overall dependability of the model.

Specificity: Accuses how good or bad the model is at producing true negatives, hence reduces false positives.

Explainability Methods:

8.1 Local Interpretable Model-Agnostic Explanations (LIME)

LIME (Local Interpretable Model-Agnostic Explanations) approximates complex models locally with simpler models, making individual predictions interpretable.

The LIME algorithm minimizes the locality-aware loss function L , represented as:

$$L(f, g, \pi_x) = \sum_{z \in Z} \pi_x(z) (f(z) - g(x'))^2, \quad (5)$$

where Z is the set of perturbed samples, $\pi_x(z)$ is the proximity weight for each sample, and $g(x')$ is the interpretable model's prediction.

8.2 Gradient-weighted Class Activation Mapping (Grad-CAM)

Grad-CAM (Gradient-weighted Class Activation Mapping) highlights important regions in an input image by using gradients from a convolutional layer.

The neuron importance weight α_{ck} for feature map A^k is computed as:

$$\alpha_{ck} = \frac{1}{Z} \sum_i \sum_j \frac{\partial y_c}{\partial A_{ij}^k}. \quad (6)$$

The Grad-CAM heatmap L_c for class c is then:

$$L_c = \text{ReLU} \sum_k \alpha_{ck} A^k. \quad (7)$$

This heatmap highlights areas crucial to the model's prediction, thus providing interpretability and aiding clinicians in understanding the model's diagnostic focus.



Fig. 7: Grad-Cam and LIME activation map using proposed model

9 COMPARATIVE ANALYSIS

9.1 Models accuracy table

Model	Accuracy
Proposed	0.9878
ResNet-152	0.7815
VGG-19	0.7091
AlexNet	0.8247

9.2 Precision, F1 Score, Recall values for All models

Model	Precision	Recall	F1-Score	Specificity	Sensitivity	Accuracy
Proposed	0.9808	0.9862	0.9834	0.9940	0.9862	0.9878
ResNet-152	0.58	0.59	0.60	0.95	0.90	0.7815
AlexNet	0.62	0.59	0.55	0.97	0.93	0.8247
VGG-19	0.78	0.70	0.67	0.99	0.92	0.7091

9.3 Proposed Model parameter Table

Parameter	Value/Description
Model Architecture	Attention-based MobileNetV2
Programming Language	Python
Deep Learning Framework	PyTorch 2.0.1
Input Image Size	224x224
Preprocessing Requirements	Image resizing, normalization
Training Epochs	25
Learning Rate	0.0001
Optimizer	Adam
Loss Function	Categorical Crossentropy

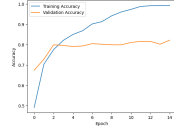


Fig. 8: Proposed Model

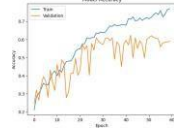


Fig. 9: ResNet-152

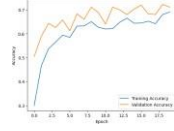


Fig. 10: VGG-19

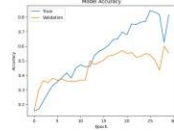


Fig. 11: AlexNet

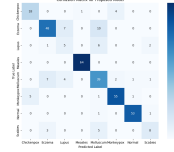


Fig. 12: Proposed Model

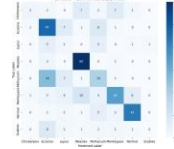


Fig. 13: ResNet-152

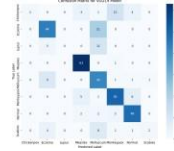


Fig. 14: VGG-19

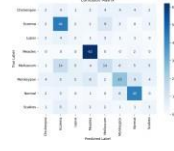


Fig. 15: AlexNet

Fig. 16: Train and Test accuracy graphs and Confusion matrix for all models

10 RESULT ANALYSIS AND CONCLUSION

10.1 RESULT AND FUTURE SCOPE

The study presented here is an attention-based MobileNetV2 model for predicting monkeypox disease. This model can reach the values of 98.78% using the EMSID dataset and even 98.19% on the original MSID dataset, thereby outperforming the traditional models like ResNet and VGG. To increase the trust of the healthcare professionals, the model further applies interpretability techniques such as Grad-CAM and LIME. This paper may suggest further extension of the dataset to more skin conditions and improve the model robustness toward real-world applications with possible deployment in clinical environments to support early diagnosis even in low-resource areas, and optimization of the model on mobile and edge devices might be promising avenues for future development.

10.2 CONCLUSION

The attention-based MobileNetV2 model resulted in high accuracy in detection, surpassing traditional models while incorporating interpretability that is healthcare trust-worthy, thus improving early diagnosis at clinical deployment, especially into low-resource areas. It is envisioned that future works would be focused on dataset extension and mobile optimization for actual application.

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